TxSuite user manual

Trionic 5

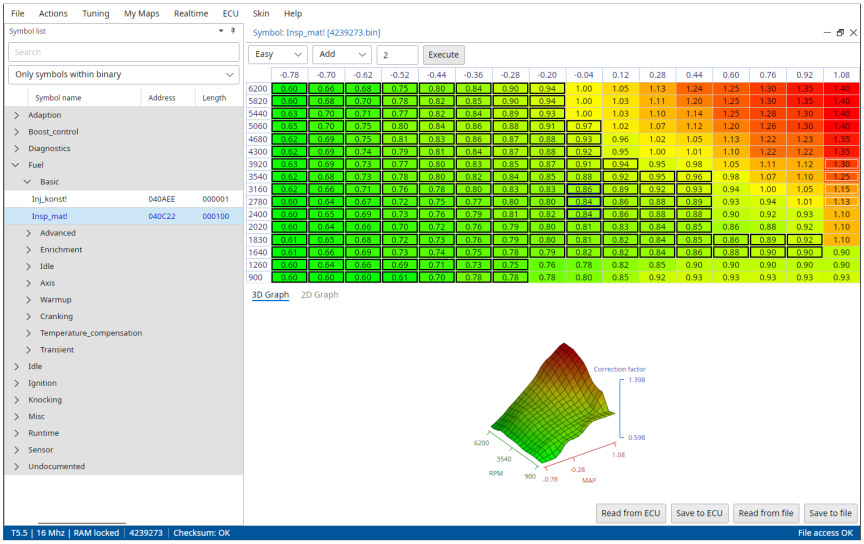


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Getting started

TxSuite is T5Suite, T7Suite and T8Suite in one program. The three suites were rebuilt on .NET 10 and Avalonia, without the commercial components the old suites needed, so TxSuite runs on Windows, Linux and macOS. Open a T5, T7 or T8 binary and TxSuite works on it the way that ECU's suite did: the same files read, edited and written, the same maps, tools and realtime panel. It talks to the ECU through TrionicCANLib, the library of the Trionic CAN Flasher.

This manual covers TxSuite with a Trionic 5 binary. Trionic 7 and Trionic 8 have a manual of their own, Help → T7 → User manual and Help → T8 → User manual.

What all three ECU's have in common is written the same way in all three manuals. The parts only your ECU has are in chapters of their own, right after the chapter they belong to.

Documentation

This manual tells you what the program does. What the ECU does, its hardware, firmware and maps and how to tune them, is in Dilemma's Trionic documentation.

For the T5 that is "Analyzing Trionic 5 with TxSuite", Help → T5 → Trionic 5 documentation.

Both are kept up to date in one place, https://github.com/roffe/TrionicDocumentation. Found something wrong or missing? Open an issue or send a PR there.

Installation

The newest TxSuite and TxExtras are on one page, https://github.com/roffe/TuningSuites/releases/latest. All downloads are on the releases page: the `Txsuite_v…` releases, and `Txsuite_nightly` with the latest build.

The stock binaries of all three ECU's are a separate download, TxExtras. TxSuite uses them wherever it needs the original file of a partnumber, Lookup partnumber for example. They go into `Binaries\T5`, `Binaries\T7` and `Binaries\T8` in TxSuite's folder.

Windows

Run TxSuiteSetup.exe. It installs to `Program Files (x86)\MattiasC\TxSuite`, next to any old T5Suite, T7Suite or T8Suite (it replaces none of them), and asks whether `.bin` files should open in TxSuite. Then TxExtrasSetup.exe puts the stock binaries into TxSuite's folder. Running the setup of a newer build over an existing installation upgrades it.

Rather not install it? TxSuite-win-x86.zip runs from wherever you unpack it, unpack TxExtras.zip into the `TxSuite` folder for the stock binaries. It doesn't open `.bin` files by itself, and its update check opens the release page instead of downloading the setup.

Linux

Download TxSuite-linux-x64.tar.gz (or -linux-arm64) and run it from where you extracted it. Unpack TxExtras.zip into the `TxSuite` folder for the stock binaries.

```
tar xzf TxSuite-linux-x64.tar.gz
./TxSuite/TxSuite
```

`./TxSuite/install-desktop.sh` adds TxSuite to the application menu and to Open with for `.bin` files, for your user only. Run it again after moving the folder.

The CombiAdapter needs libusb and the udev rule next to the program:

```
sudo cp TxSuite/70-txsuite.rules /etc/udev/rules.d/
sudo udevadm control --reload && sudo udevadm trigger
```

There is also an AppImage, all of TxSuite in one file (TxSuite-linux-x64.AppImage, or -linux-arm64). `chmod +x` it and start it. It needs FUSE, which most desktops have; without it start it with `--appimage-extract-and-run`. The stock binaries go next to the AppImage, unpack TxExtras.zip in its folder. The udev rule isn't in the AppImage, take `70-txsuite.rules` from the tar.gz.

macOS

Download TxSuite-osx-arm64.zip (Apple silicon) or -osx-x64 (Intel). The build is not signed, so clear the quarantine flag after extracting it, then start it from Terminal or by double-clicking TxSuite in Finder:

```
unzip TxSuite-osx-arm64.zip
xattr -dr com.apple.quarantine TxSuite
./TxSuite/TxSuite
```

Supported interfaces

TxSuite supports the same interfaces as the Trionic CAN Flasher. On a T5 it talks to the ECU's P-bus.

Interface	**Windows**	**Linux**	**macOS**
Lawicel CANUSB	Lawicel CANUSB DLL driver	FTDI serial port	FTDI serial port
CombiAdapter	libusb (WinUSB driver)	libusb + udev rule	libusb
OBDLink SX / EX, ELM327	serial port	serial port	serial port
Just4Trionic	serial port	serial port	serial port
SLCAN	serial port	serial port	serial port
Kvaser HS	Kvaser CANlib driver	Kvaser linuxcan	not supported
J2534 (Beta)	vendor DLL (32-bit)	vendor `.so`	not supported
Wideband lambda (realtime)	serial port	serial port	serial port

The adapters still need their own drivers, see Adapter setup in the flasher's README, https://github.com/roffe/Trionic#adapter-setup. Which adapter TxSuite uses is set in File → Settings → Common.

The P&E micro and USB BDM programmers of T5Suite are gone, there is no driver for them on .NET 10. Reading and flashing goes over CAN, see the ECU menu.

Settings and log files

Settings and log files are in `%APPDATA%\MattiasC\TxSuite` on Windows and `~/.config/MattiasC/TxSuite` on Linux and macOS. File → Settings → Logging decides which log files get written, at what level and for how long, and its Open button shows the folder on any system.

The first start takes over the settings of T5Suite, T7Suite and T8Suite once, with their Recent lists merged into one. After that TxSuite keeps its own settings and the old suites are left alone.

Coming from the old suites

TxSuite looks and works like the old suites, with some differences:

- One window for all three ECU's. The file you open decides the ECU.
- One menu bar instead of the ribbon. Want the ribbon back? File → Settings → Common → Show ribbon bar instead of menus.
- One settings dialog with a tab per ECU.
- Excel export and import are CSV now, and the map tables copy as text that pastes into Excel.
- Autologging, sound notifications, full screen and the panel snapshot (F9) are gone.
- New are the symbol and command palettes, colour blind modes, a dark skin, dropping files on the window and multi-step undo in the map viewer.

Updates

TxSuite looks for a newer release when it starts and from Help → Check for updates, and shows the result in the status bar. When there is one it asks. OK downloads the setup on Windows and opens the release page elsewhere (and for the Windows zip).

Disclaimer

This is Open Source software that pokes around in your car's control system. The authors of the tools shall not be held accountable for how you decide to use the tools. If you are not careful, you can easily brick your car with these tools so please use this software with care.

The main window

TxSuite is controlled by the menu and the symbol list shown on the left hand side of the screen. Maps, compare results, the realtime panel and the other tools open as windows in the workspace on the right. The status bar at the bottom shows what is open and how it's doing.

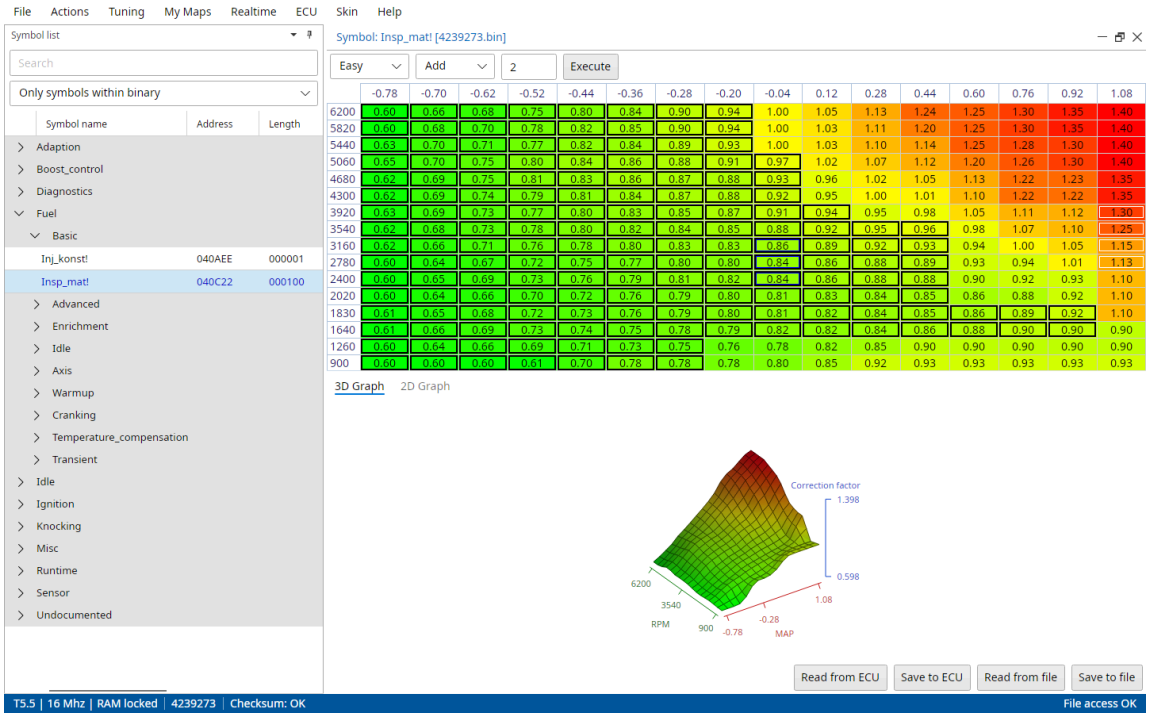


Image 1: The main window with a Trionic 5 fuel map open

The menu

The menus are File, Actions, Tuning, My Maps, Realtime, ECU, Skin and Help. What every ECU has is always there. An ECU's own items only show while a file of that ECU is open, in the place they had in that ECU's old suite. With no file open only the shared items show.

Items that need an open file are greyed out until you open one. The ECU follows the file as well: Connect ECU, Read ECU and the other ECU actions need a file of that ECU open, because the file tells TxSuite which ECU it is talking to. Without one TxSuite says "A file must be loaded first: it tells which ECU (Trionic 5, 7 or 8) to connect to."

The menus are described one by one in the chapters after this one.

The ribbon bar

The old suites had a ribbon bar instead of menus. File → Settings → Common → Show ribbon bar instead of menus brings it back, at once.

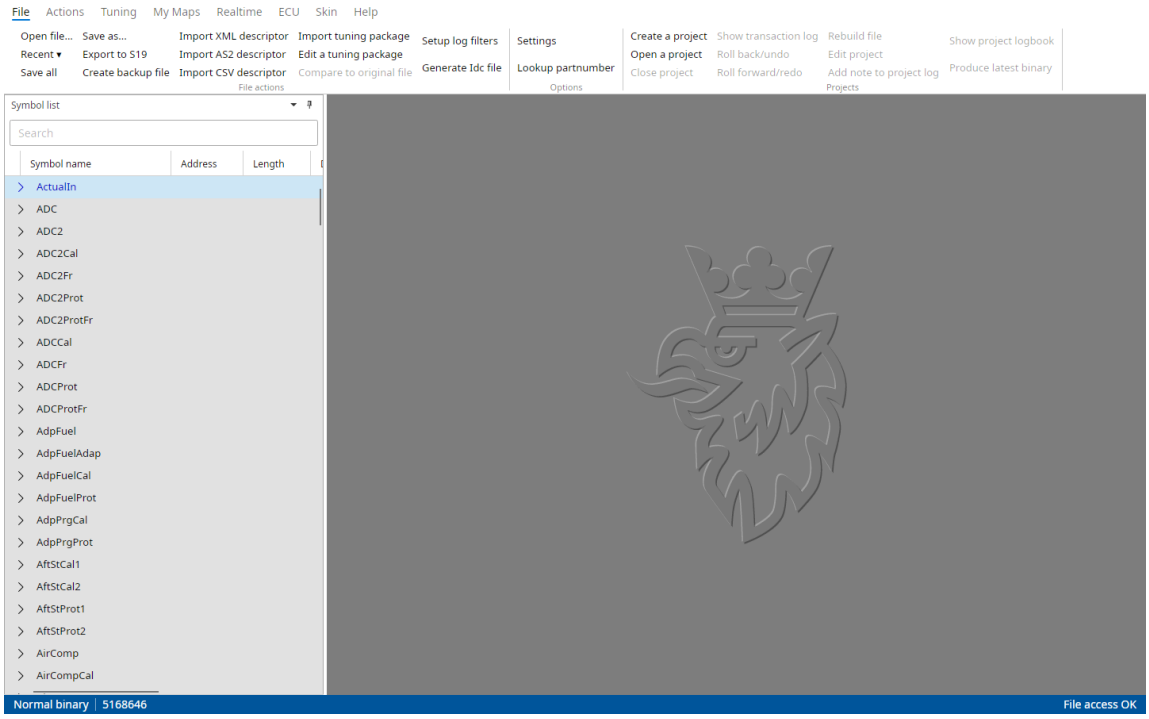


Image 2: The ribbon bar, File tab

The ribbon is made from the menus, so it always has the same items. Each menu is a tab, each section of a menu a group with its items as buttons. Groups that don't fit in the window fold into a button that drops them down.

- Double-click a group's caption to fold it, double-click a folded one to keep it open.
- Drag a group by its caption to put it somewhere else on its tab.
- Right-click a caption and pick Reset this tab's groups to get the menu's order back.

The folds and order are remembered per ECU and per tab. The buttons have text only, no icons, and there is no Exit button: close the window.

Opening a file

File → Open file… opens a binary or a Motorola S19 file. TxSuite looks at the file and picks the ECU by it: T5 and T7 by size and content, T8 by its 1 MB size.

A file opens from these places too:

- File → Recent, the files opened before.
- A file dropped on the main window.
- Open with in the file manager, or the file name on the command line.
- Opening a project, or Lookup partnumber's Open this file and Create new from original.
- File → Settings → Common → Auto load last file on startup reopens the last file (or project) when TxSuite starts.

Opening a file of another ECU closes everything of the one before, as when the suites were separate programs. Maps with unsaved changes are asked about first, Cancel keeps it all. Nothing switches while an ECU is being flashed or a T5 autotune runs. A file no ECU takes gives "File is not a Trionic 5, 7 or 8 binary file!".

After opening, the symbol list on the left is filled with the symbols of the file, see the next chapter.

Dropping files

Files dropped on the main window load like they do from the menu:

- a binary opens as with File → Open file,
- an SRAM dump loads as with Actions → Import SRAM snapshot,
- a logfile of the open file's ECU opens as with Realtime → Load logfile.

Everything else (tuning packages, descriptors, CSV files, reports) stays with its menu item, so nothing changes the binary by accident. A file dropped on a hex view opens in that hex view.

Windows in the workspace

Maps, compare results, the hex view, the realtime panel and the other tools open as windows inside the workspace. Move them by their title bar, resize them by their edges, and minimize, maximize or close them with the buttons on the right of the title bar. A double-click on the title bar maximizes a window too.

Drag a window's title bar out of the main window and it floats in a window of its own, handy with a second screen. Its "Dock back into main window" button brings it back.

The Skin menu has the window layout: Cascade windows, Tile windows horizontally, Tile windows vertically and Dock all floating windows.

When a compare shows a map of two files, both share one window with a tab per file.

The status bar

The status bar shows, from left to right:

- "Open/dev binary" for open (development) software, else "Normal binary",
- the file name and "Checksum: OK" or what is wrong with it,
- the SRAM snapshot that is loaded, if any,
- the connection to the ECU,
- the result of the update check.

On a T5 it starts with the ECU type, CPU speed and RAM lock of the file instead, as in "T5.5 | 16 Mhz | RAM locked".

The right side shows the progress of what is running (reading, flashing, extracting) and whether the file can be written, "File access OK".

Symbol palette and command palette

Like in VS Code, Ctrl+P opens a symbol by any part of its name and Ctrl+Shift+P runs any menu command (Cmd on macOS). Type a part of what you are looking for, move with the arrow keys, Enter or a click picks, Escape closes.

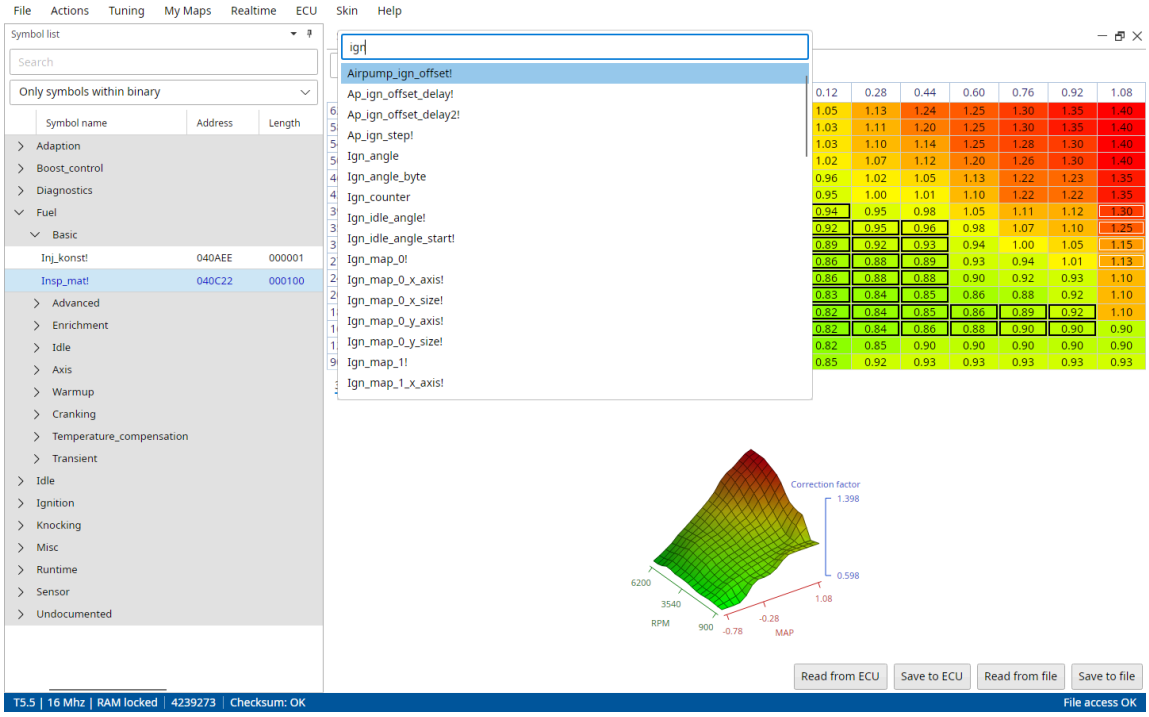


Image 3: Ctrl+P, the symbol palette

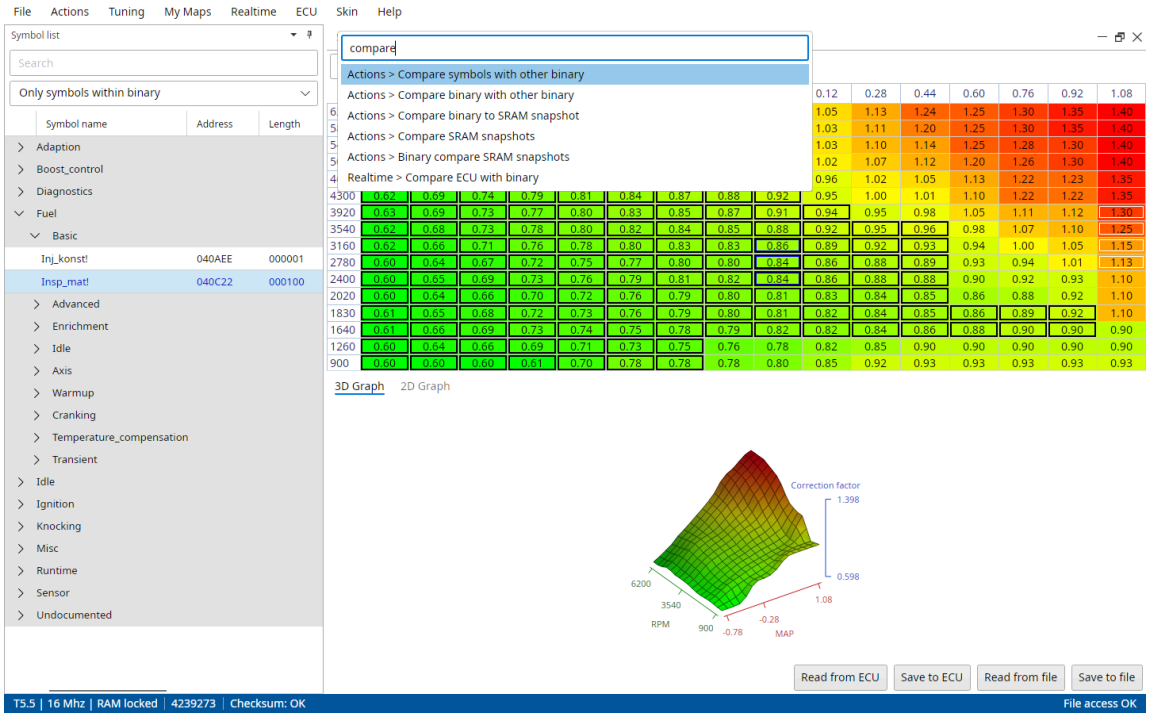


Image 4: Ctrl+Shift+P, the command palette

The symbol palette lists the symbols of the open file that a double-click in the symbol list would open. The command palette lists every menu command that is available right now as “Menu > Item”, the Recent files and the map entries of the Tuning and My Maps menus included.

Keyboard shortcuts

Key	What it does
Ctrl+P	Open a symbol by name
Ctrl+Shift+P	Run a menu command
Shift+F1	Realtime → Toggle realtime panel
F6	Realtime → Write log marker
Enter	Open the selected symbol in the symbol list

Shift+F5 is Realtime → Toggle Autotune, with File → Settings → T5 → Advanced mode enabled.

The map viewer has its own keys, see Working with maps.

The symbol list

After a file is opened the symbol list on the left hand side of the main window is filled with its symbols. The symbols are grouped by category and every category starts folded, click the arrow in front of one to open it. The Left arrow key folds the category of the selected symbol again.

TxSuite reads the symbol table from the file. A T5 binary always has one, so there is nothing to extract or import.

The columns are Symbol name, Address, Length, Description and User description. More are hidden, like Number, SRAM address and Category, the column chooser shows them (see Sorting and grouping below). Addresses and lengths are hexadecimal or decimal by File → Settings → Common → Show addresses and lengths in Hex.

The User description column is yours: click a cell to type a note for that symbol. It is saved next to the bin in `<bin>.xml`.

With File → Settings → Common → Show descriptions under symbol names each symbol's description shows on a line of its own under its name, as the old T7Suite and T8Suite did. File → Settings → Common → Symbol list font size makes the list's text larger or smaller.

Selecting symbols

Choose a symbol by double-clicking it or by selecting it and pressing Enter. A new window opens in the workspace with the map of that symbol, titled "Symbol: <name> [<file>]". Opening a map that is open already brings its window to the front. Working with that window is the next chapter.

It also works the other way around: the map in front is selected in the symbol list, with its category opened.

Ctrl+P is the quick way, type any part of a symbol's name and Enter opens it.

Searching

Type in the Search box above the list and it shows only the symbols with that text in their name, description, user description, address or length. The categories open so the matches show. Clear the box and the categories fold again, except the one of the selected symbol.

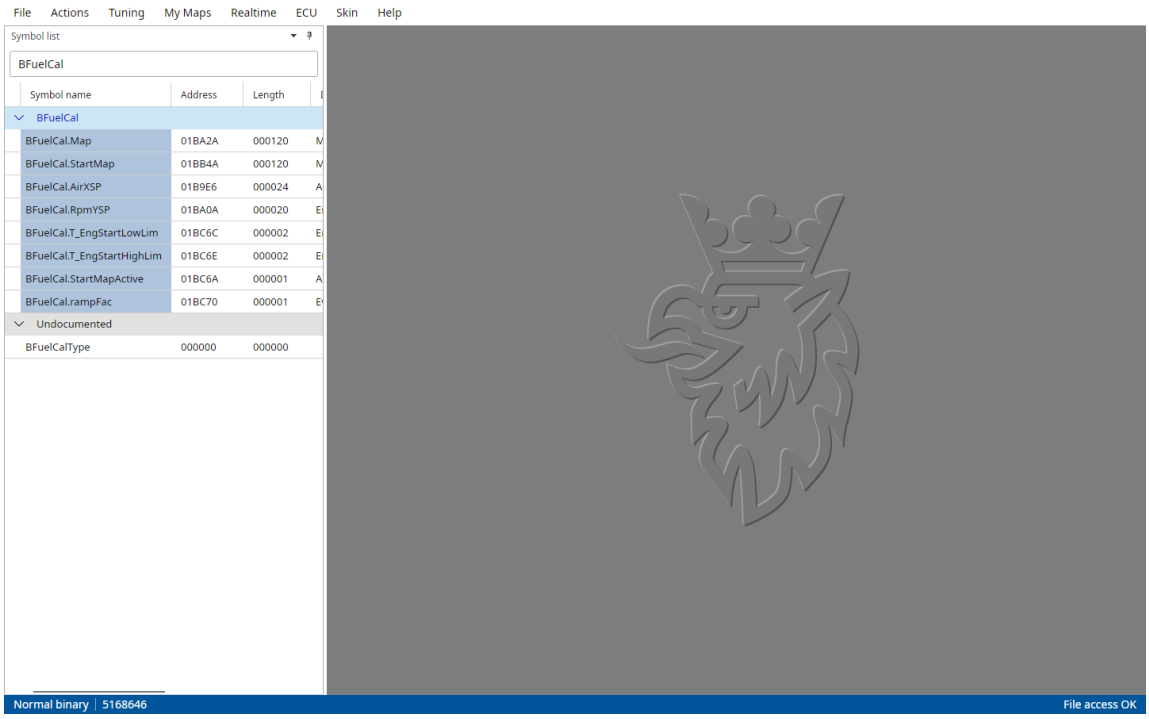


Image 5: Searching the symbol list

Filtering

Below the Search box is a filter. By default it shows “Only symbols within binary”, the symbols that are actually stored in the flash. Choose “All symbols” to see the symbols that only live in SRAM too. TxSuite switches to all symbols by itself when you connect to the ECU or load an SRAM snapshot.

Right-click a column header and choose Show auto filter row for a filter box per column. Text typed in one shows only the symbols with that text in that column, and several boxes work together.

Sorting and grouping

Click a column header to sort on that column, click it again to sort the other way around. The categories stay in their order, the symbols are sorted within them.

Right-click a column header for more:

- Sort ascending, Sort descending and Clear sorting.
- Group by this column, Group by category (the default) and No grouping.
- Hide this column, and the Column chooser to show or hide any column.
- Best fit and Best fit (all columns) to size the columns to their content.
- Show auto filter row, see Filtering.

Columns can be dragged to another place and resized by their edges.

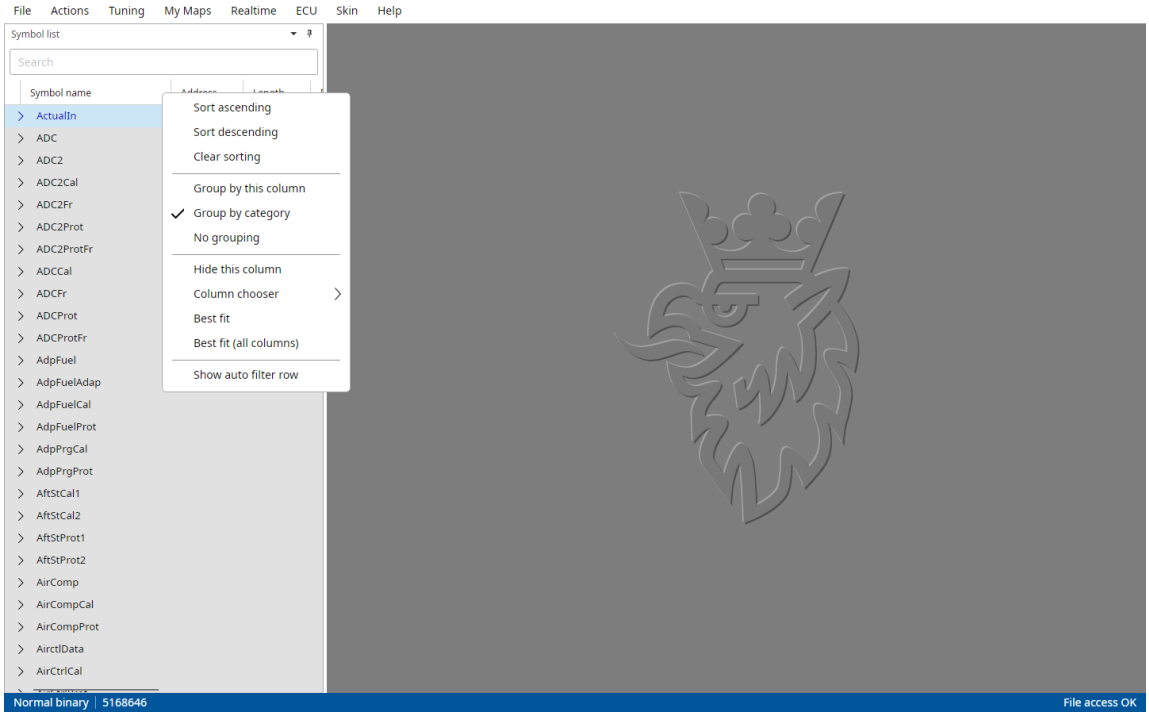


Image 6: The column header menu of the symbol list

Map preview popup

Point at a symbol's name and its map pops up as a read-only table next to it. File → Settings → Common → Show map preview popup turns it off and on.

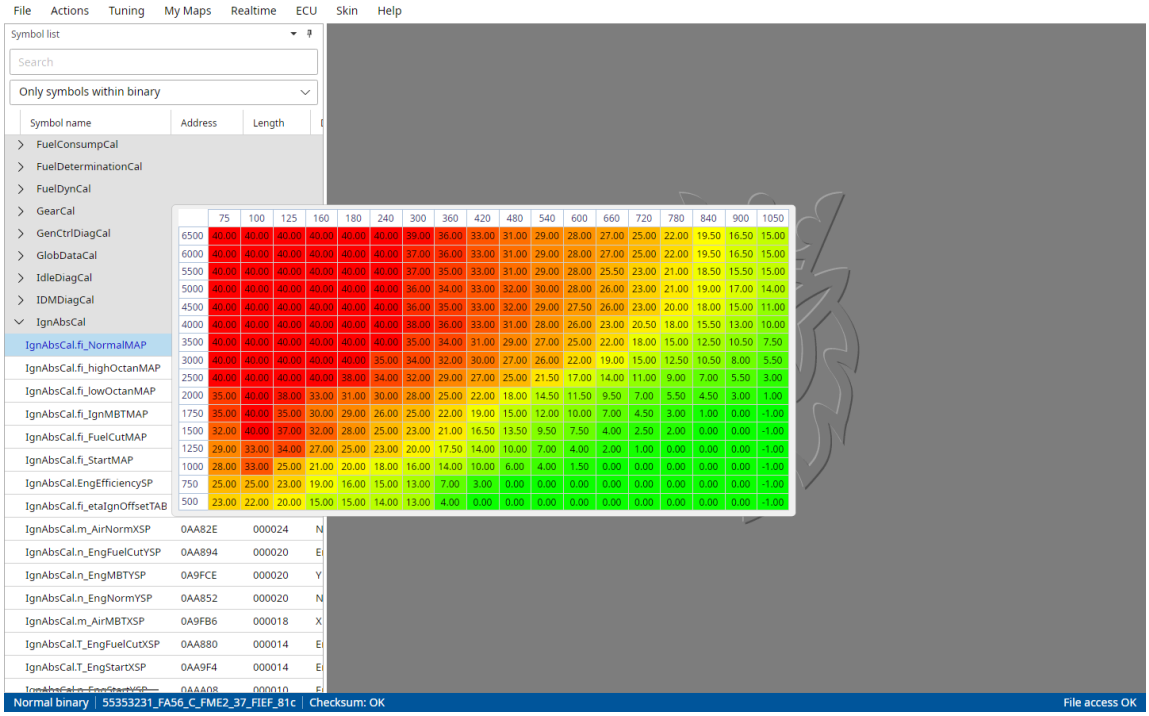


Image 7: The map preview popup of a T8 ignition map

The symbol menu

Right-click a symbol for these:

Add to realtime list
Adds the symbol to the realtime panel's free logging list.

Add to MyMaps
Adds the symbol to your own My Maps menu.

Read from SRAM file
Opens the symbol's map from the SRAM snapshot that is loaded with Actions → Import SRAM snapshot.

Browse axis information
Shows the symbol in Actions → Browse axis information.

Export symbollist as CSV
Saves the whole symbol list as a CSV file.

Add to realtime user maps
Adds the map to the User maps tab of the realtime panel.

Hiding the symbol list

File → Settings → Common → Hide symbol window hides the symbol list to a tab at the side of the window. The tab brings the list out again, and it slides away when you open a map. The pin button at the top of the list does the same.

Working with maps

A symbol opened from the symbol list shows in a map viewer, a window in the workspace with the table on top and its graphs below. You can see one in the picture at the start of The main window chapter.

From top to bottom the map viewer has:

- the toolbar: the view type, and the math on the selection (operation, value and Execute),
- the table, with the x axis above it and the y axis on its left,
- the graphs, a 3D Graph tab and a 2D Graph tab,
- the buttons Read from ECU, Save to ECU, Read from file and Save to file.

The splitter between the table and the graphs moves up and down. File → Settings → Common → Show graphs in mapviewer turns the graphs off, for slower computers. Read from ECU and Save to ECU only show for symbols the ECU has in its RAM.

Hexadecimal mode

The view type box in the toolbar switches the table between Hex, Decimal, Easy and ASCII.

Hex
: The raw values in hexadecimal.

Decimal
: The raw values in decimal, as the ECU stores them.

Easy
: The values converted with the map's factor and offset to what they mean, degrees, bar, mg/c and so on. Easy view is best if you don't know what this means ;)

ASCII
: The bytes as text, read only.

The view type a map opens in is set by File → Settings → Common → Default view type for maps. Each map viewer can still be switched to another view type.

Color indicators

All maps are displayed in colours from green (low values) through yellow to red (high values), over the smallest and largest value of that map. If you find this hard to tell apart, File → Settings → Common → Color blind mode has other colour scales: Universal, Protanopia, Tritanopia and Deuteranomaly. They are used for the table, both graphs and the airmass result viewer, open map viewers change at once.

Maps opened from the ECU while connected keep their own colours, so you can see they show what's live in the ECU.

File → Settings → Common → Mapviewer cell spacing puts space between the cells, 0 to 20 pixels. With a space the cells show as separate tiles without the lines between them.

Some T5 maps have marks in their cells:

- a black rectangle around the cells that operate in closed loop (Insp_mat!, Ign_map_0!, Fuel_knock_mat!),
- a padlock on cells below the knock pressure limit and on cells locked by autotune,
- a blue rectangle for the knock adaption range and a white one for the boost adaption range,
- a rectangle on the cells of the AFR feedback maps that have measurements.

Selecting cells

Click a cell to select it, drag to select a block. Shift+click extends the selection, Ctrl+click adds or removes cells. The arrow keys move the selection, Ctrl+A selects the whole map.

Adjusting values in a map

To avoid that you have to adjust all values of a “large” map manually some features have been added to the map editor:

Key	What it does to the selected cells
Plus	adds 1
Minus	subtracts 1
PageUp	adds 10 (0x10 in hex view)
PageDown	subtracts 10 (0x10 in hex view)
Home	sets them to the maximum value
End	sets them to 0

Typing a value starts editing, F2 edits the focused cell starting from its value, Enter stores it and Esc throws it away. With several cells selected the typed value goes into all of them.

The toolbar does math on the selection: pick Add, Multiply, Divide or Fill, type the value and click Execute.

Ctrl+Z undoes the last change and Ctrl+Y redoes it, as many steps back as you like. Undo and Redo are in the table’s right-click menu too.

Copy and paste

To be able to get your work done faster you can select one or more cells in a table and copy them to the clipboard by right-clicking and selecting “Copy selected cells”. With nothing selected the whole map is copied.

To paste the cells select the location where you want the cells to appear (this could be in another map and even in another binary), right-click and select “Paste selected cells” and then “At original position” or “At currently selected location”.

Copy selected cells uses TxSuite’s own format, for pasting into maps. Ctrl+C copies the selected cells as plain text with tabs, that pastes straight into a spreadsheet.

Smoothing

Select a row, a column or a block and pick Smooth selection in the table’s right-click menu. A row or column becomes a straight line between its first and last value, a block gets the average of its neighbours in each cell inside it, the cells on its edge stay as they are.

Editing the axes

Edit x-axis and Edit y-axis in the table’s right-click menu open the axis as a map of its own, to change its values.

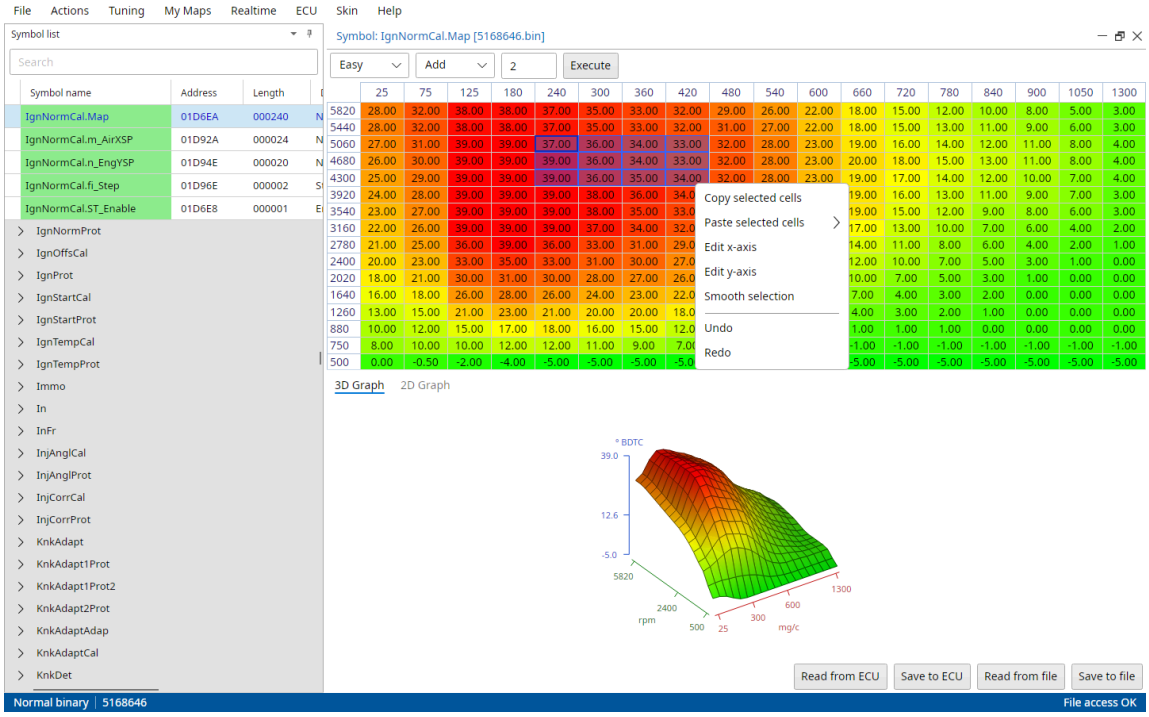


Image 8: The table's right-click menu over a block of cells

Graphs

The 3D Graph tab shows the map as a surface. Drag with the left mouse button to turn it, with the right button to roll it and with the middle button to move it, and the mouse wheel zooms.

The 2D Graph tab shows one column of the map as a line over the y axis, the slider below it picks which column.

With File → Settings → Common → Synchronize mapviewers on, viewers of the same map follow each other: selecting cells in one selects them in the others, and turning one graph turns the others.

Saving changes

Save to file writes the map into the binary and updates the checksum. Read from file throws away the changes and reads the map from the file again.

Closing a viewer with changes asks "Data was mutated, do you want to save these changes in you binary?", Yes saves, No throws the changes away and Cancel keeps the viewer open. Closing TxSuite asks the same for each changed viewer. File → Save all saves every open viewer at once.

In a project every save is a transaction in the project's log, see the Projects chapter.

Maps from the ECU

When the ECU is connected, Read from ECU reads the map from the ECU's RAM and Save to ECU writes it there, every viewer of that map follows. When the ECU refuses a write TxSuite says so.

On a T5 every map opens with the ECU's SRAM data while connected, and Save writes the ECU and the file in one go. The ECU only takes changes with RAM unlocked, see RAM locked in the Firmware information chapter. A map that only lives in SRAM opens from the loaded SRAM snapshot when you're offline, otherwise TxSuite says "Symbol resides in SRAM and you are in offline mode...".

File → Settings → Common → Auto update SRAM viewers every … seconds reads these maps from the ECU again and again while connected. This enables you to keep an eye on for example the knock map. Viewers with unsaved changes are left alone. Don’t set the interval too short, it takes time from the realtime panel.

Checksum

For you to be able to use the binary after altering maps or other information within it, the checksum must be updated after alteration of data. TxSuite does that every time it writes a map into the binary.

File → Settings → Common → Auto update checksum makes sure the checksum is right everywhere else too. Advised is to keep this turned on! With it off TxSuite asks before it corrects a wrong checksum, and before it flashes a file with one.

Actions → Verify checksum checks the open file’s checksum by hand.

File menu

The File menu opens, saves, imports and exports files, and holds Settings, Lookup partnumber and the Project submenu.

File → Open file…

Lets you choose a binary file on your computer, or a Motorola S19 file which is converted first. TxSuite picks the ECU by the file and extracts all the relevant information from it, see Opening a file in The main window chapter.

File → Recent

Shows you a list of recently opened files for easy access, of all three ECU's in one list.

File → Save all

Normally TxSuite saves a map when you close its viewer and say Yes. If you have map viewers with changes open you can save all of them to the file at once with this option.

File → Save as…

This lets you choose a different location and/or filename for the binary file you currently have open. Then TxSuite asks whether to open the copy, Yes and you work on the copy.

File → Create backup file

Lets you create a backup file for the binary at this point. It is wise to create backups before you start to make big changes to your file. If a project is opened, the backup file will be stored within the project folder, otherwise it will be stored in the folder where the bin file is located.

File → Compare to original file

If you have a file opened and TxSuite has the original file it was made from, this option will be enabled. It allows quick access to the changes that were made to the opened binary file, the result is the same list as Actions → Compare symbols with other binary. The original files are the stock binaries from TxExtras, in TxSuite's `Binaries` folder.

File → Setup log filters

The realtime panel logs everything it reads. With log filters you tell TxSuite which lines of a log to keep, for example only the lines above a certain rpm, so you only look at what's relevant.

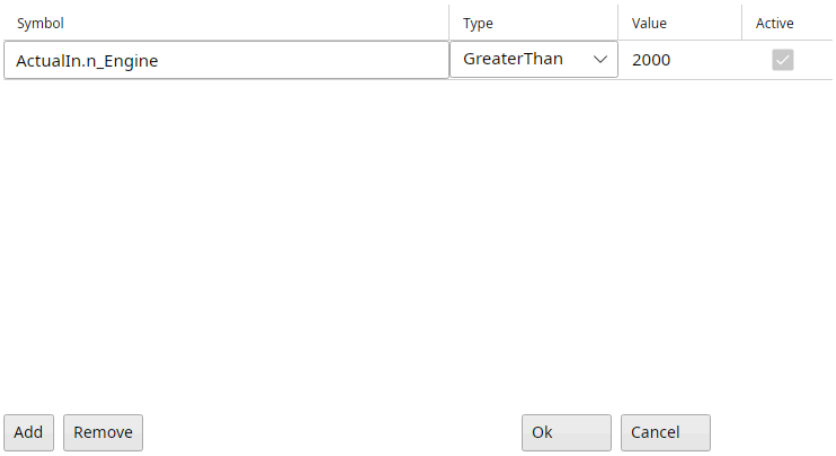


Image 9: File → Setup log filters with one filter

Each filter has a symbol, a type (GreaterThan, SmallerThan or Equals), a value and whether it's active. A line is left out when an active filter doesn't hold for it. The filters are applied in the log viewer and in both log exports, the logfile itself keeps everything.

File → Generate Idc file

Writes an IDC script with the symbol names of the open file, for disassembling the binary in IDA Pro. If you're not sure what that means, you don't need this.

File → Settings

This option lets you alter the settings that control the behaviour of the application. They have a chapter of their own, Settings.

File → Lookup partnumber

Lets you look up a partnumber and the stock binary that belongs to it. It finds the partnumbers of all three ECU's, whichever file is open.

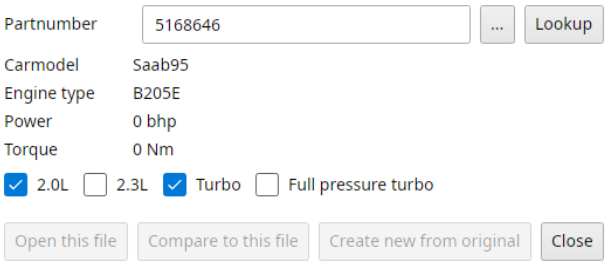


Image 10: File → Lookup partnumber, 5168646 looked up

In this screen you can either enter a known partnumber and click Lookup, or click the "..." button to browse all known partnumbers. TxSuite shows the car model and engine type the partnumber belongs to, and for a T5 or T7 partnumber the power and torque too.

When the stock binary is in TxSuite's `Binaries` folder (TxExtras) you can use it:

Open this file
: Opens the stock binary.

Compare to this file
Compares the open file to the stock binary, only for a partnumber of the open file's ECU.

Create new from original
Saves a copy of the stock binary under a name you choose and opens it, a clean start for a new tune.

The "…" button shows every known partnumber with a tab per ECU, the open file's ECU selected. The list is grouped by car model and engine type, every group starts folded. The partnumbers that have a stock binary in the `Binaries` folder are coloured. Double-click a row, or select it and click Ok, to look it up. Export saves the list as tab separated text.

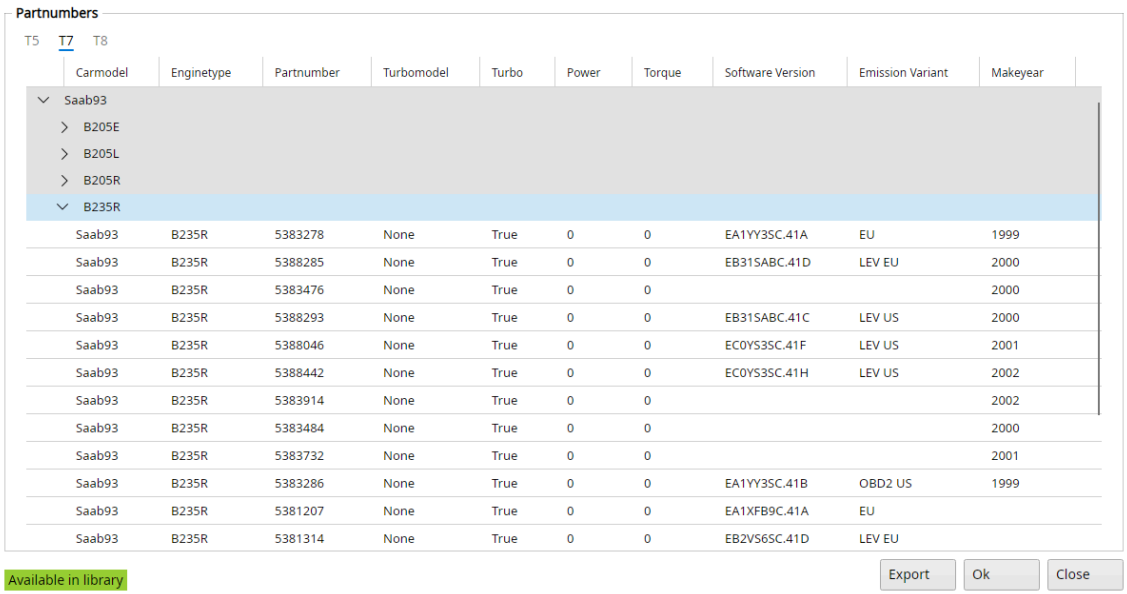


Image 11: The partnumber list, T7 tab

File → User library

Your own library of T5 binaries next to the stock ones. Add files scans a folder for T5 bins and lists each with what TxSuite guesses about it: tuning stage, injectors, map sensor, torque, E85, T7 valve, partnumber, software, CPU speed and RAM lock. Open selected opens a file, Compare to selected compares it to the open file, and Clear library empties the list. The library is kept in `UserLib.json` in the `T5` folder of TxSuite's settings folder.

File → Project

The Project submenu lets you work on your tune in a project, with roll-back and roll-forward, versions and a logbook. It has a chapter of its own, Projects.

File → Exit

Come on Einstein, what would that do?

Closing TxSuite asks about maps with unsaved changes first, and it won't close while an ECU is being read or flashed.

Projects

TxSuite allows you to work on your tune in a project based fashion. Projects have the advantage of having roll-back and roll-forward functions, having versioning and keeping things tidy and together. You are advised to work in a project if you do more than simple easy-tune your binary file.

A project is a folder with a copy of its binary, the transaction log of every change made to it, a logbook and the backups. The projects are in File → Settings → Common → Project folder, the same folder for all three ECU's. Every time a project opens TxSuite makes a backup of its binary first.

All project items are in the File → Project submenu.

File → Project → Create a project

This option lets you create a new project from the binary you have open. The dialog is filled in from the open file, the binary is copied into the new project folder.

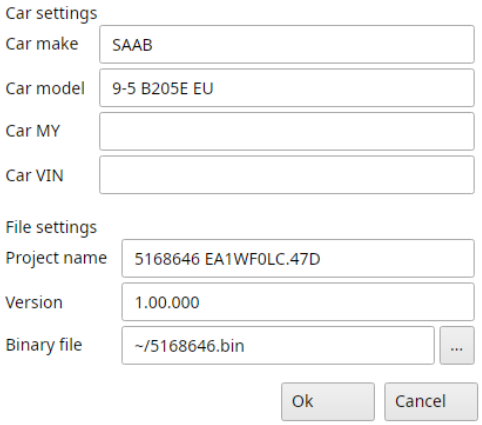


Image 12: Create a project, filled in from the open file

File → Project → Open a project

Lets you select and open a previously created project. The list shows each project's number of backups and transactions, when it was last changed and its version. If no projects are available (none were created before) TxSuite will notify you with a message box.

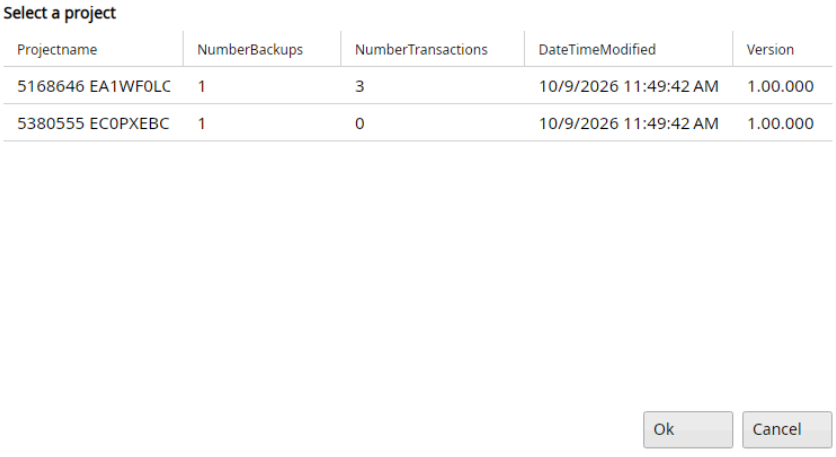


Image 13: Open a project

Opening a plain file closes the project. With File → Settings → Common → Auto load last file on startup the last project reopens when TxSuite starts.

File → Project → Close project

Closes the project and allows you to work on single files again.

File → Project → Show transaction log

Shows the transaction log for the current project. Since a project contains only one binary file, you could see this as an undo/redo list for the project binary file. Each row is one write to the binary: when, which symbol, your note, the address and length, and whether it was rolled back. You can roll back or roll forward any entry from here, and type a note for it.

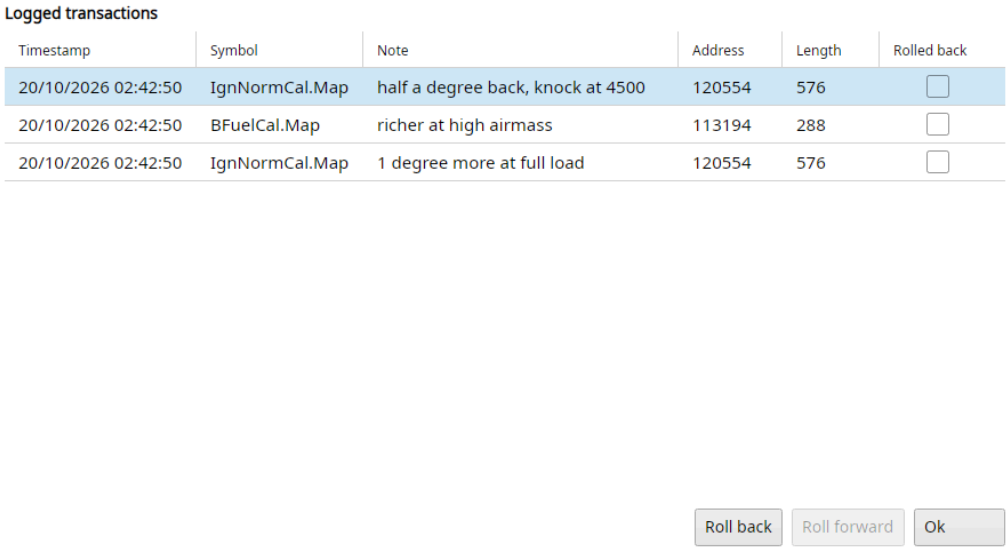


Image 14: The transaction log after three map saves

When the log grows past 2000 entries TxSuite offers to purge it.

With File → Settings → Common → Request project notes on, TxSuite asks you for a note every time a change is saved in a project. This way, you can keep track of changes very easily.

File → Project → Roll back/undo

Lets you rollback the last transaction made to the binary file. If no transactions are available to rollback, the item will be disabled.

File → Project → Roll forward/redo

Lets you roll forward (redo) the last transaction that was undone/rolled back. If no transactions are available to roll forward, the item will be disabled.

File → Project → Rebuild file

This enables you to rebuild a project file (binary) up to a certain point in time. TxSuite will ask you for a date and it will restore (if possible) the file that you had at that specific date. Store result as current project file puts it in the project, otherwise TxSuite asks where to save it.

File → Project → Edit project

Lets you edit the properties of the open project. This enables you to manage your version numbering yourself. Version numbers are stored, together with all other project properties, in the xml file in the project folder. Renaming the project renames its folder too.

File → Project → Add note to project log

You can add a note to your project with a timestamp attached to it, so you can keep track of changes you made in your setup. For example, if you start using different injectors or mount a bigger intercooler, you can enter a note of this into the project log for later reference.

File → Project → Show project logbook

Shows you in detail what has happened in your project.

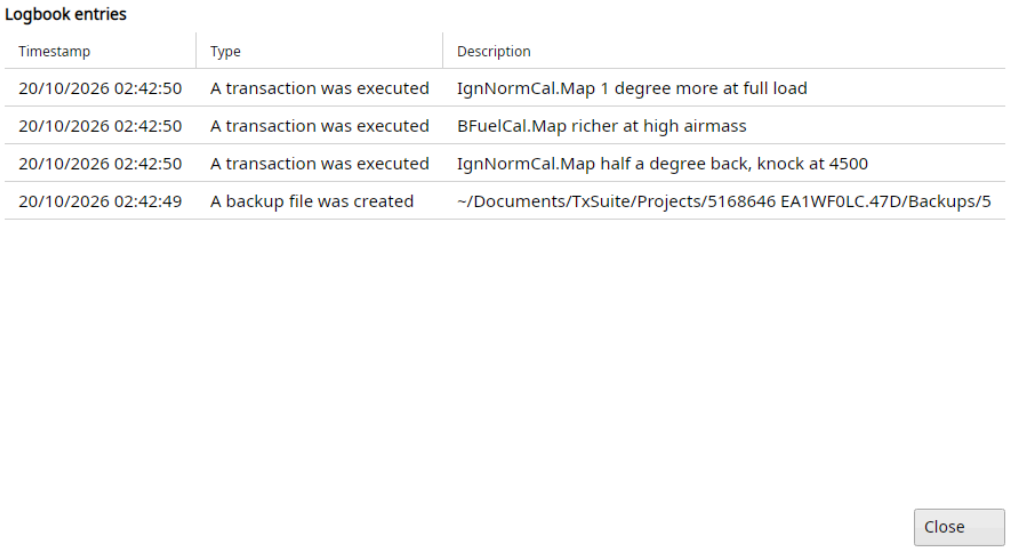


Image 15: The project logbook

File → Project → Produce latest binary

Lets you export the project binary in its current state so you can save it in another location easily.

Settings

File → Settings opens one dialog for every ECU, with the tabs Common, Logging, T5, T7 and T8. Common and Logging hold what every ECU has, each ECU's tab the options only that ECU has. Ok applies every tab at once, also to a file that is open.

Hover over an option for a tip on what it does.

This chapter has the Common and Logging tabs, the T5 tab is in the next chapter. The T7 and T8 tabs are in their manuals.

Settings → Common

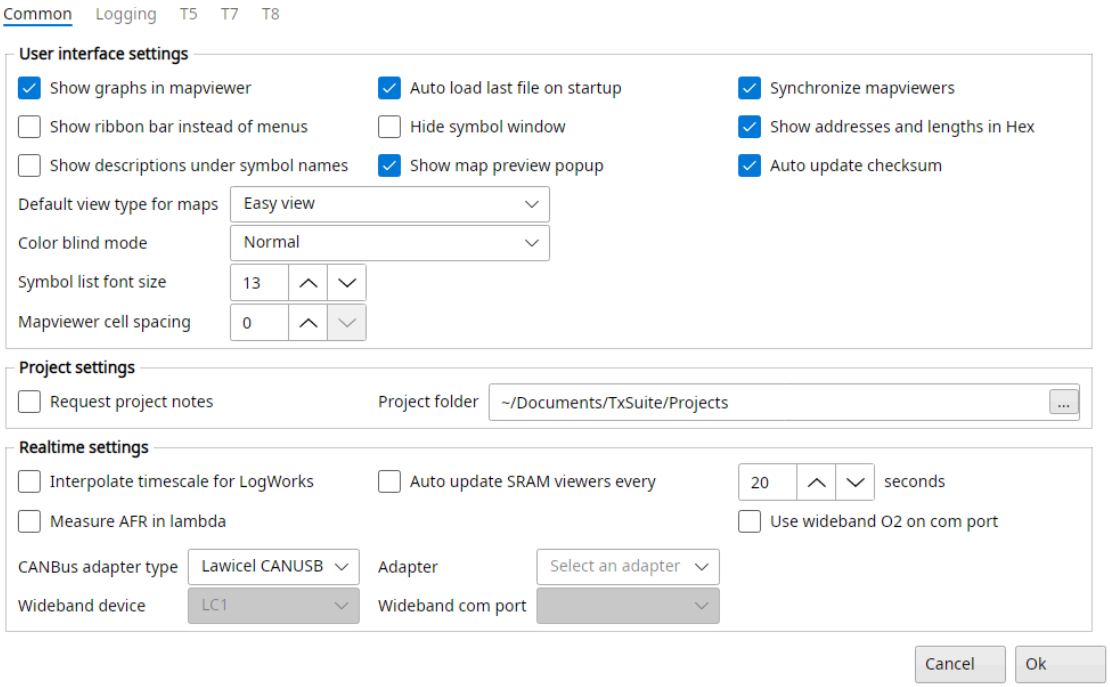


Image 16: File → Settings, the Common tab

User interface settings

Show graphs in mapviewer
To gain performance in lightweight computers you can switch off the graphical display in the map viewer altogether with this option.

Show ribbon bar instead of menus
Shows the menus as a ribbon bar at the top of the window, as in the old suites, see The main window.

Show descriptions under symbol names
The symbol list shows each symbol's description on a line of its own under its name, as the old T7Suite and T8Suite did.

Auto load last file on startup
If you work on one file for a prolonged period of time you might want TxSuite to reopen the last file automatically when you start it. If the last opened item was a project, TxSuite will reopen that project for you at start-up.

Hide symbol window
To gain more space on the main screen you can hide the symbol list to a tab at the side of the window. The tab brings the list out again, and it slides away when you open a map.

Show map preview popup
Pointing at a symbol's name in the symbol list shows its map as a read-only table in a popup.

Synchronize mapviewers
Map viewers that show the same map follow each other: selecting cells or turning the 3D graph in one does the same in the others.

Show addresses and lengths in Hex
TxSuite can display addresses and lengths of symbols in decimal form or in hexadecimal form. This option allows you to switch between the two, in the symbol list, the search results and the compare results.

Auto update checksum
Having this checked will rid you of the constant worry whether or not the checksum of your file is valid. Advised is to keep this turned on! With it off TxSuite asks before it corrects a wrong checksum, and before it flashes a file with one. Map saves always update it.

Default view type for maps
The view type a map opens in: Hexadecimal view, Decimal view or Easy view. Easy view is best if you don't know what this means ;)

Color blind mode
A colour scale for maps and their graphs that is easier to tell apart with colour blindness: Normal, Universal, Protanopia, Tritanopia or Deuteranomaly.

Symbol list font size
The size of the text in the symbol list, 8 to 24 pixels.

Mapviewer cell spacing
Space between the cells of a map table, 0 to 20 pixels. At 0 they touch as in the old suites.

Project settings

Request project notes
If you are working in a project and you save changes to the binary file, TxSuite will prompt you for a note to describe the change made. If disabled, the prompting will be skipped.

Project folder
Lets you decide where to store projects (the project root folder), for every ECU. You can use this if you want all the projects to be located on a separate disk partition for example.

Realtime settings

Interpolate timescale for LogWorks
LogWorks uses a fixed interval and the frame rate TxSuite reaches differs most of the time. Check this option and TxSuite will interpolate the values to match the interval LogWorks works with, when exporting a logfile to LogWorks.

Measure AFR in lambda
If you prefer AFR measurements in lambda values instead of AFR values, enable this.

Auto update SRAM viewers every … seconds
If you have map viewers open that show the ECU's RAM, TxSuite can refresh them every 5 to 60 seconds while connected. This enables you to keep an eye on for example the knock map.

Use wideband O2 on com port
Reads a wideband controller on a serial port while the realtime panel runs. Its reading is used as the realtime AFR and is shown and logged as Wideband. It overrides the wideband through an ECU symbol on the ECU's tab.

CANBus adapter type
Lets you choose which CAN bus interface you have: Lawicel CANUSB, Kvaser HS, CombiAdapter, OBDLink SX, OBDLink EX, J2534, Just4Trionic, SLCAN or ELM327.

Adapter
Which adapter of that type to use. For an OBDLink, ELM327, Just4Trionic or SLCAN adapter this is its serial port. The CombiAdapter is found by itself, so the list stays empty for it.

Serial speed
The speed of the serial connection to an ELM327 or a Just4Trionic. It shows for SLCAN too, but SLCAN adapters always use 3000000 baud. The OBDLink SX and EX pick their speed themselves.

Wideband device
The type of wideband controller on the wideband com port: PLX, LM1, LC1, LM2, ZT2, AEM, STAG or LambdaShield.

Wideband com port
The serial port the wideband controller is connected to.

Settings → Logging

TxSuite writes log files of what it does, a file per log and per day. They are what to post when something doesn't work.

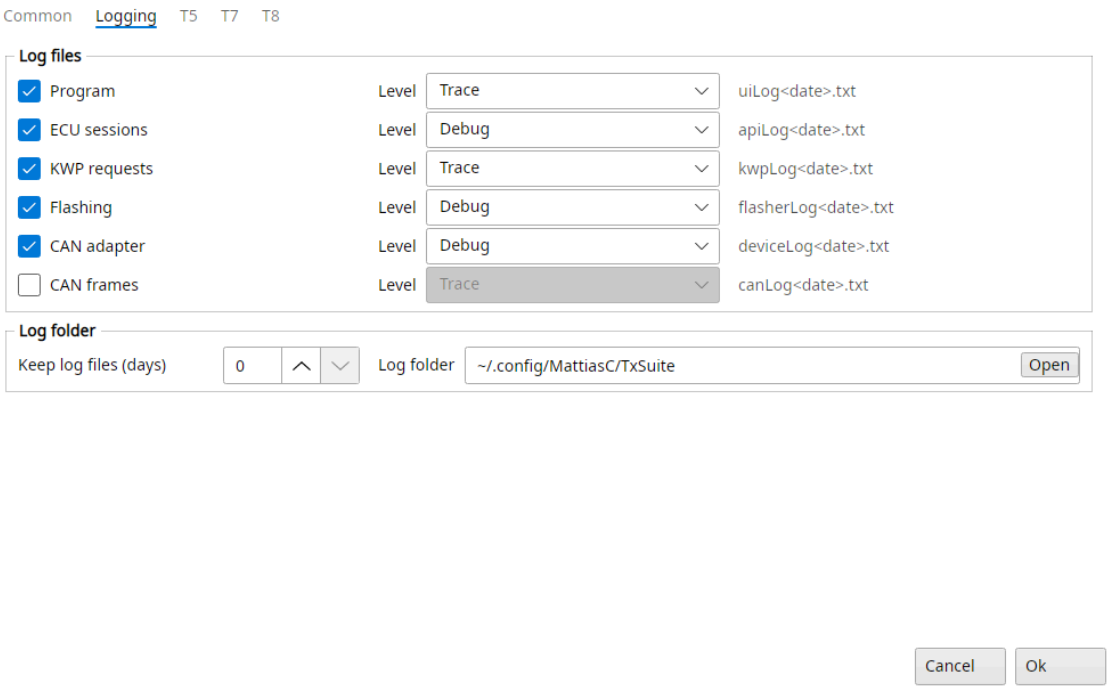


Image 17: File → Settings, the Logging tab

Each log can be turned on or off and gets a level of its own. Trace logs everything, Debug a little less, then Info, Warn and Error. A level also logs every one after it, so Error logs the errors alone.

Program
What TxSuite and its T5, T7 and T8 parts do: opening and parsing binaries, symbols, checksums, settings, the realtime panel.

ECU sessions
The T5, T7 and T8 sessions: connecting, security access, reading and writing memory, fault codes.

KWP requests
Every KWP2000 request to a T7 and its answer.

Flashing
Reading and flashing an ECU's flash, block by block.

CAN adapter
The adapter's driver: opening it, its settings, the frames it couldn't send, its errors.

CAN frames
Every CAN frame sent and received. It grows fast and slows the communication down, for finding communication problems only. Off by default, it is the old suites' Enable canbus logging.

Keep log files (days)
Log files last written longer ago than this many days are deleted when TxSuite starts and when the settings are saved. 0 keeps them all.

Log folder
Where the log files are, its Open button opens the folder.

Trionic 5 settings

The T5 tab of File → Settings has the options only a T5 has.

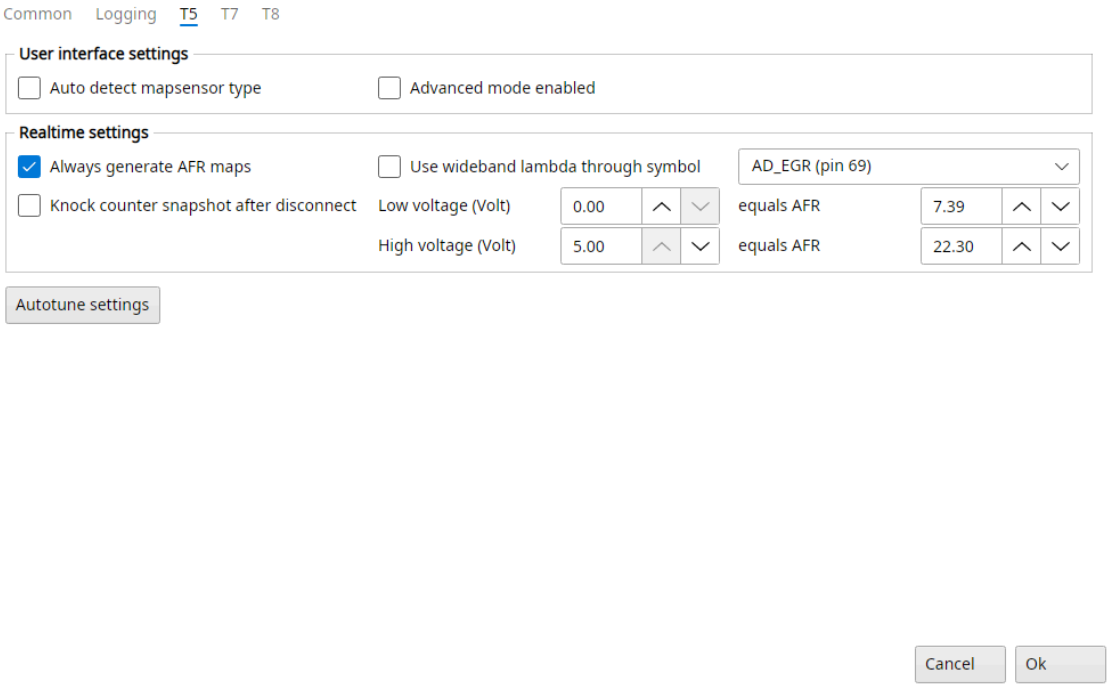


Image 18: File → Settings, the T5 tab

Settings → T5

User interface settings

Auto detect mapsensor type
: Most pro made tunes don't have the marker for the correct mapsensor type in the binary file. This option lets TxSuite auto detect the correct mapsensor type for the current binary file by examining certain maps.

Advanced mode enabled
: This will switch TxSuite into advanced mode for a T5. This means you will have access to more options: the advanced tuning wizards, autotune and the advanced tabs of the realtime panel. If you are not sure whether you need this, leave it unchecked.

Realtime settings

Always generate AFR maps
: TxSuite sets up the AFR maps (target, feedback and counter maps in an `AFRMaps` folder next to the file) as soon as a file is opened, instead of when they are first needed. Users without a realtime connection will never need the AFR maps of course.

Knock counter snapshot after disconnect
: Saves the ECU's knock counter map (Knock_count_map) as a `.KNK` file in the Snapshots folder (the project's, else next to the file) when you stop the realtime panel or disconnect from the ECU. T5.5 only, Realtime → Knock map snapshots lists them.

Use wideband lambda through symbol
TxSuite allows you to setup your engine with a wideband lambda sensor like the LC1 from Innovate Motorsports. The wideband controller can deliver a linear analogue signal to the ECU, read through AD_EGR or AD_cat. We advise you to use AD_EGR for that (pin 69).

Low voltage, High voltage, equals AFR
Settings to calibrate the wideband lambda controller input: the low voltage equals the AFR next to it, the high voltage the other one, and voltages in between convert on a straight line.

A wideband controller on a serial port is set up on the Common tab instead, Use wideband O2 on com port.

Settings → T5 → Autotune settings

Lets you choose options for the Autotune fuel and Autotune ignition functions. Please note that you need to be running a wideband lambda sensor to be able to use Autotune fuel.

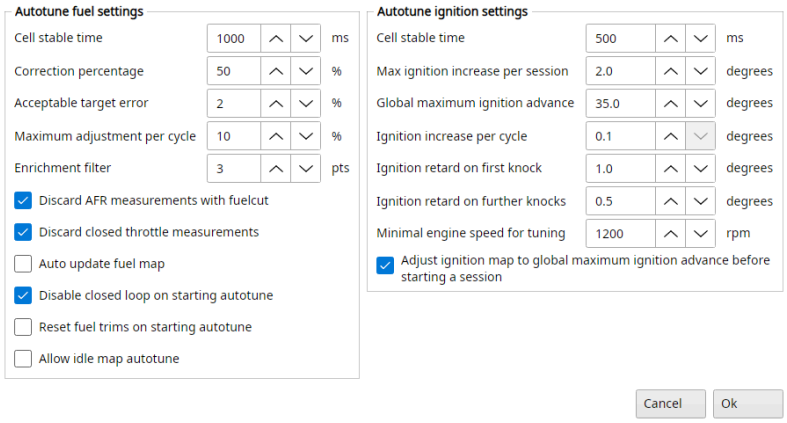


Image 19: File → Settings → T5 → Autotune settings

Autotune fuel settings

Cell stable time
Determines how long (in milliseconds) the system should reside in a certain load/rpm cell before TxSuite validates the AFR measurement.

Correction percentage
Determines how much a certain AFR error will be corrected. Setting this to 50% (default) means 50% of the established AFR error will be corrected for each session.

Acceptable target error
Determines how big the AFR error is allowed to be without TxSuite taking action in that certain load/rpm cell.

Maximum adjustment per cycle
Determines how big the maximum adjustment will be in percentages measured from the original fuel map.

Enrichment filter
Determines how big enrichment/enleanment factors are allowed to be while still doing adjustments to the fuel map.

Discard AFR measurements with fuelcut
Determines whether measurements while the system is in fuel cut will be ignored.

Discard closed throttle measurements
Determines whether measurements with closed throttle plate will be discarded.

Auto update fuel map
This option decides the mode in which Autotune works. If this is checked the system will try to map the fuelling entirely on its own. No feedback from the user is requested. After an Autotune session the user can however revert to the previous fuel map if he/she chooses to do so. If this is not checked the Autotune function will be more like a question & answer game between TxSuite and the user. After every session a map is prompted in which TxSuite shows which cells are to be altered to reach the requested AFR. The user can select/deselect cells to his/her liking and update the fuel map with the selected cells.

Disable closed loop on starting autotune
Some advanced users have closed loop operation turned off completely. In that case you might want to prevent TxSuite to disable closed loop operation when starting an autotune session.

Reset fuel trims on starting autotune
Starts the session with the ECU's fuel adaption reset, so it doesn't mix in.

Allow idle map autotune
Lets Autotune fuel work on the idle fuel map too.

Autotune ignition settings

Cell stable time
How long (in milliseconds) the engine has to stay in a cell before TxSuite advances or retards it.

Max ignition increase per session
How many degrees a cell may be advanced in one session at most.

Global maximum ignition advance
No cell is advanced past this.

Ignition increase per cycle
How many degrees a cell is advanced each time the engine has been in it long enough without knock.

Ignition retard on first knock and on further knocks
How many degrees a cell is retarded when it knocks, the first time and after that. A cell that knocked is locked, it is never advanced again.

Minimal engine speed for tuning
Below this rpm nothing is tuned.

Adjust ignition map to global maximum ignition advance before starting a session
Caps every cell of the ignition map at the global maximum first.

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Actions menu

The Actions menu has the checksum, the information about the open file, the compare and transfer tools and the SRAM snapshots. Its last section, the tools, is different for every ECU and is described in the chapters after this one.

Actions → Verify checksum

Checks the checksum of the open file and shows the result in the status bar, “Checksum: OK” or “Checksum: invalid”. This will not be needed if the auto checksum update function is turned on.

Actions → Firmware information

Lets you view and modify a lot of parameters in the ECU, T5Suite’s Trionic options. It has a chapter of its own, Firmware information.

Actions → VIN decoder

Lets you decode SAAB VIN numbers into readable format, from car model and engine type to gearbox, and checks the VIN’s check digit.

A T5 file holds no VIN, so it opens empty.

Type a VIN and press Enter or click Decode.

The picture is the stock T7 binary 5168646, it carries a placeholder VIN instead of a car’s. That’s why it decodes as Unknown with a checksum error.

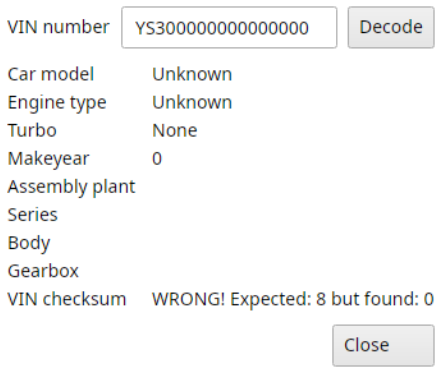


Image 20: Actions → VIN decoder

Actions → Browse axis information

Shows you the relations between maps and axis symbols. This is very informative to browse and learn. Double-click a row to open its map, or the axis you clicked on. The symbol menu’s Browse axis information opens it at that symbol.

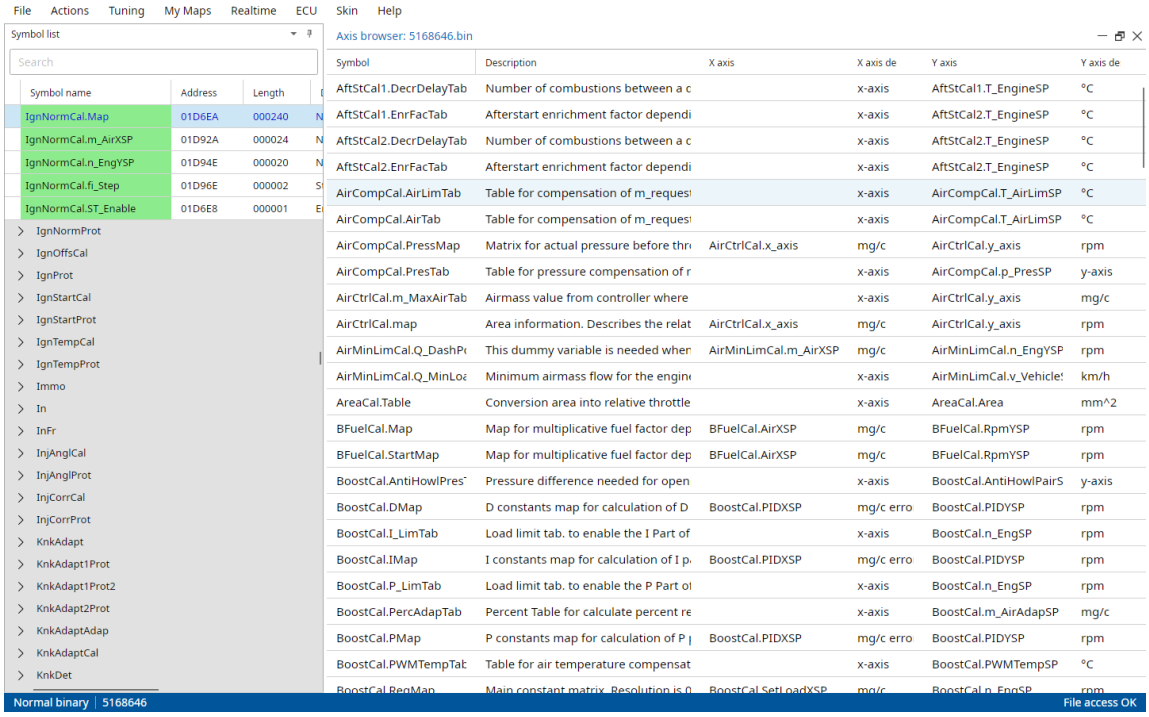


Image 21: Actions → Browse axis information

Actions → Show interrupt vectors

This is a little tool for techies.

It shows the start-up vector information for the Motorola MC68332 microprocessor.

Actions → Show disassembly

Lets you disassemble the binary file into human readable text format also known as assembler language. If you're not sure what that means, you don't need this.

The listing is saved next to the bin as `<bin>.asm` and opens in an editor with search and replace. Next to it are the bin's bytes, the cursor in one follows the other. When the `.asm` file already exists TxSuite asks whether to make it again.

Actions → Compare symbols with other binary

The most time-consuming thing is comparing maps for different firmware versions by hand. TxSuite gives you the tools to compare maps in different binaries with a click of the mouse. Comparing binary files (actually the maps inside them) is a very educational activity, you will see which maps are different in for example B234L and B234R engine types.

To do this you must first open the primary binary you want to compare. Then, select Compare symbols with other binary and select the secondary binary you wish to compare the first one with. TxSuite will now display a list of symbols that differ in the selected binaries, grouped by category. The list shows how many values differ, the percentage and the average difference. Symbols that only one of the files has are coloured and say in which file they're missing.

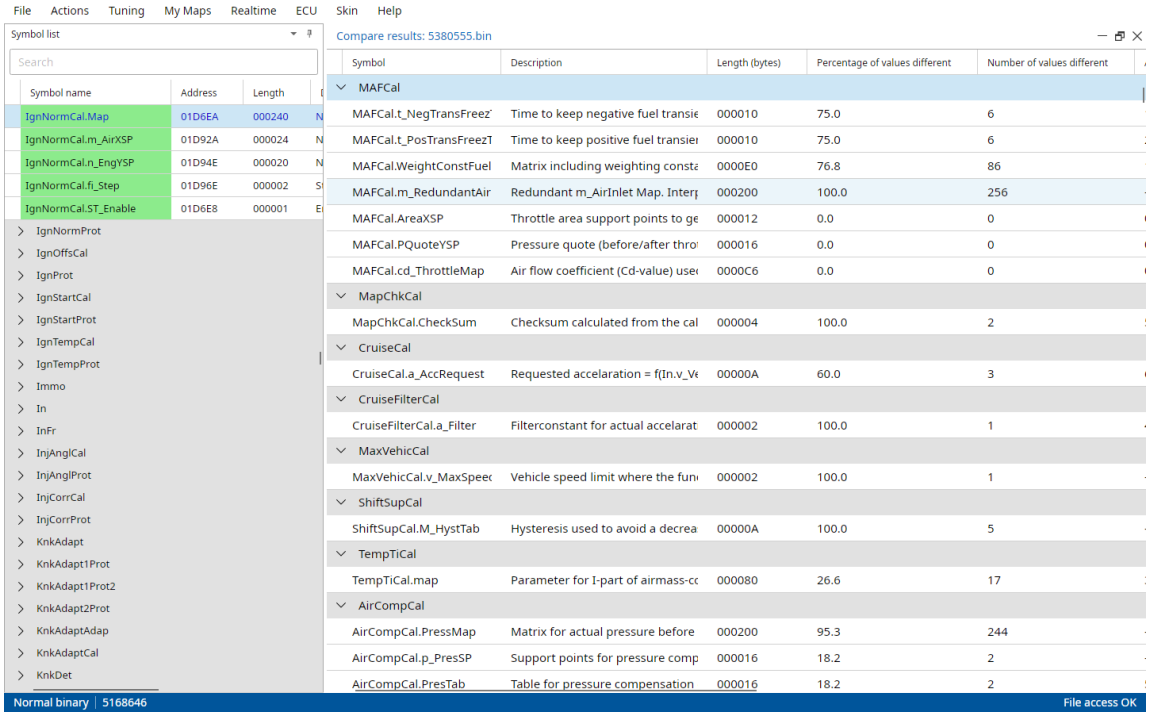


Image 22: Compare results against another T7 binary

Double-clicking a symbol in this result list opens its map from both files, in one window with a tab per file. Right-click a symbol for more:

Open
: The same as a double-click.

Show differences map
: One map with the difference between the two files.

Export to CSV
: Saves the result list as a CSV file.

You can select several files to compare to. TxSuite then shows a Compare list first, with the number of differing symbols per file, Enter or a double-click shows the results of that file.

Actions → Compare binary with other binary

Lets you do a binary (byte-by-byte) compare of two files. This is a good tool to verify whether a programming session was successful or not. Sequences of steps would be:

1. Program ECU
2. Read file from ECU
3. Compare original and downloaded file with Compare binary with other binary.

If it says 0 lines differ the files are identical (successful programming).

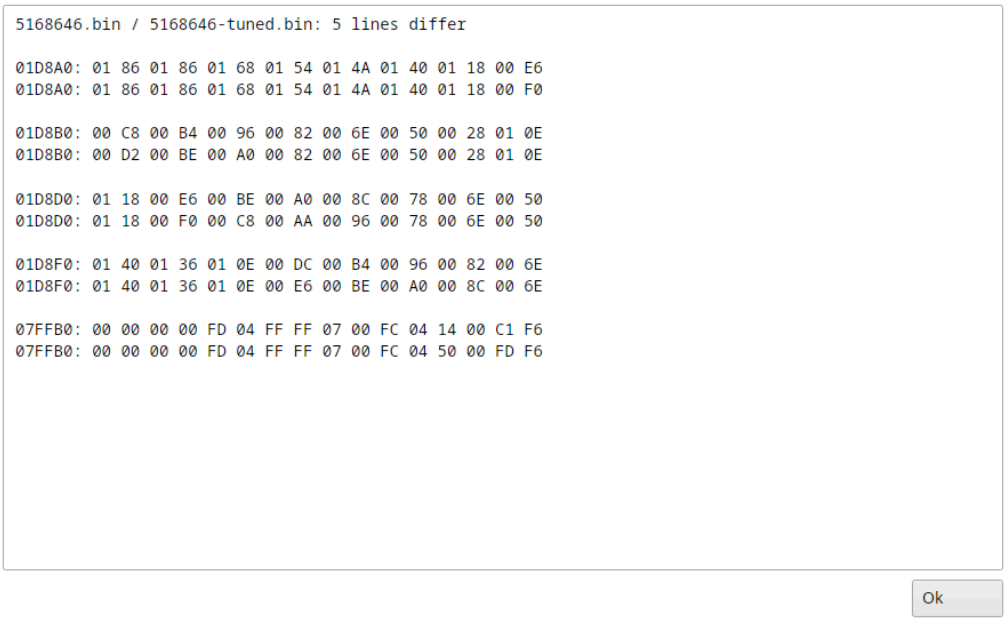


Image 23: Compare binary with other binary

Actions → Transfer maps to another binary

If you have a set of tuned maps in one binary file and you want to have them transferred to a different base file, you can use this function. The wizard asks for the target file and shows a list of symbols to choose from, it remembers your last choice. A backup of the target is made before anything is written, and a report shows what was done.

Please note that different base files might have different map sizes and TxSuite might be unable to transfer them. Exchanging data between T5.5 and T5.2 and v.v. is very dangerous!

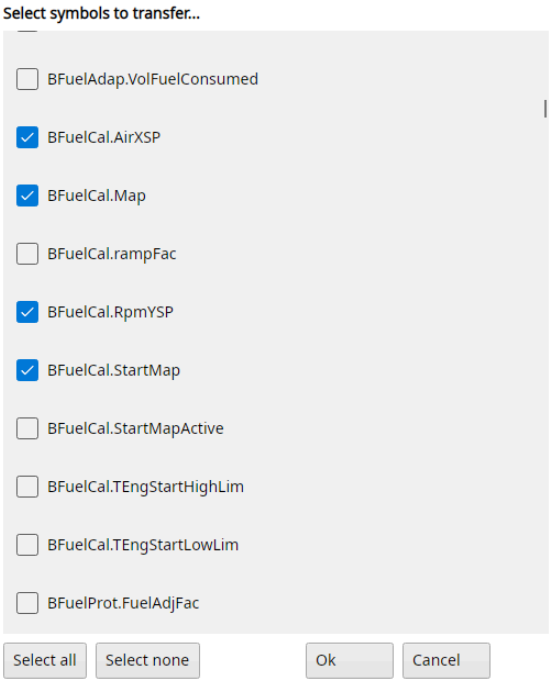


Image 24: Selecting the symbols to transfer

Actions → Search map content

Allows you to search maps for a specific value, a number (as shown in Easy view) or a text. It can search the symbol names and descriptions too, and only maps of a given length. The result will be shown in a list, double-click a row to open its map.

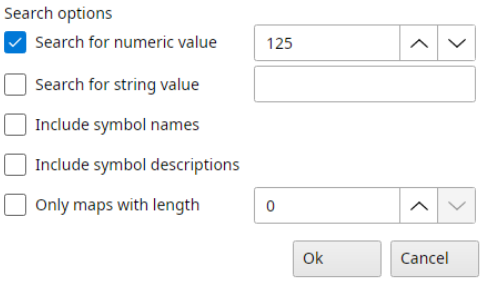


Image 25: Search map content

Actions → Export map to CSV

Saves the map selected in the symbol list to a CSV file, which opens in Excel or any other spreadsheet. It replaces the old suites' Excel export. The file is named `<bin>~<symbol>.csv`.

Actions → Import map from CSV

Reads a map back from a CSV file, the other way around. A file named `<bin>~<symbol>.csv` goes into that symbol after asking, otherwise into the map selected in the symbol list.

Actions → View file in hex

Lets you explore the binary file in hexadecimal form (advanced users!) The symbol under the cursor is shown. You can change bytes and save them in place, TxSuite makes a backup copy first and asks about unsaved changes when you close it. A loaded SRAM snapshot can be viewed in hex as well.

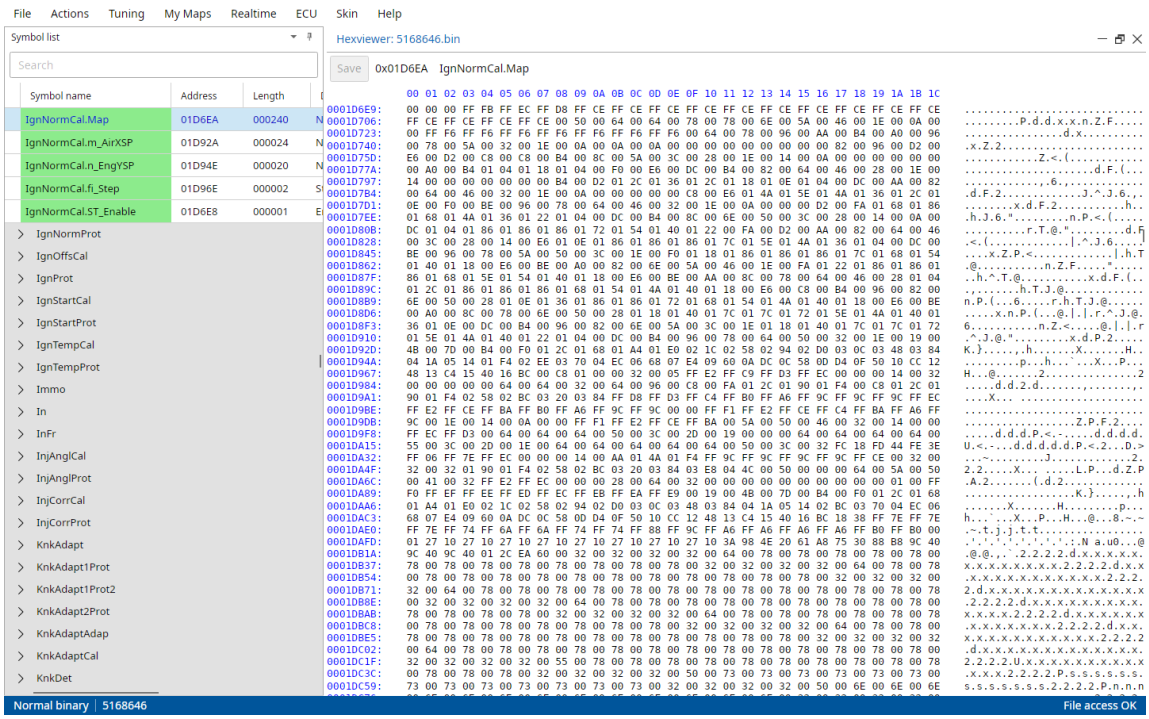


Image 26: Actions → View file in hex

SRAM snapshots

An SRAM snapshot is a dump of the ECU's working memory (`.RAM`). It holds the maps that only live in RAM and what the ECU learned (adaption), and while you tune live the changes that aren't in the file yet. ECU → Get SRAM snapshot makes one.

Actions → Import SRAM snapshot

Loads an SRAM snapshot for the open file. Symbols that only live in SRAM then open from it, and the symbol menu's Read from SRAM file shows a map as it was in the snapshot. The status bar shows the loaded snapshot.

Actions → Compare binary to SRAM snapshot

Compares the open binary file to an SRAM snapshot and shows what the ECU has learned (adapted) so far OR which changes you've made in realtime to the ECU which are still stored in SRAM. The result is a difference list from which you can choose to view the separate maps.

Actions → Compare SRAM snapshots

Lets you compare two different SRAM snapshots from the ECU to see which settings were altered, either by you or the ECU itself (adaption). The result is a difference list from which you can choose to view the separate maps.

Actions → Binary compare SRAM snapshots

Mostly for exploration of SRAM dump files. You can use this function to find the byte differences in two SRAM files.

Actions → Import SRAM snapshot into binary

This allows you to import the most important adaption data into the binary file from an SRAM dump that you made from the ECU after it has been allowed to adapt. You can choose which items to import (use with caution, advanced users only!)

Firmware information

Actions → Firmware information, T5Suite's Trionic options, allows you to view and modify a lot of parameters in the ECU. They are stored in the binary: the flags of the program mode (Pgm_mod!) and the markers TxSuite keeps in the file's footer. Ok writes them, as transactions in a project, and updates the checksum. Cancel leaves the file alone.

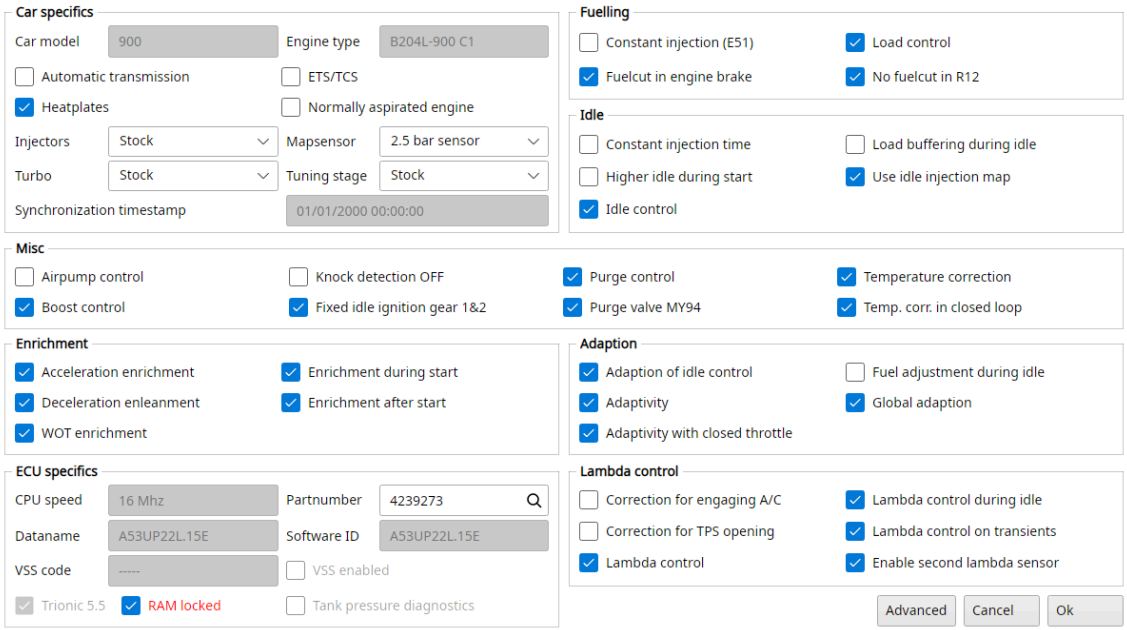


Image 27: Actions → Firmware information on a T5

Opening the Pgm_mod! symbol from the symbol list opens this window too. Options the binary doesn't have are greyed out.

Car specifics

Car model
Information field, showing the car model.

Engine type
Information field, showing the engine type.

Automatic transmission
Sets the car in automatic transmission mode or in manual transmission mode. This should match the gearbox you have in your car.

ETS/TCS
Turns the Electronic Throttle System on/off. This should match your car. If the car is equipped with TCS, turn it on, otherwise turn it off.

Heatplates
Some models are equipped with heat plates, located between the head and the intake manifold.

Normally aspirated engine
Indicates whether the car has a turbocharger mounted or not.

Injectors
Information field used as an indicator to TxSuite which injectors you have mounted in the car: Stock, Green giants, Siemens 630cc/min (60lb/h), Siemens 875cc/min (80lb/h) or Siemens 1000cc/min.

Mapsensor
Information field used as an indicator to TxSuite which mapsensor type you are using: 2.5 bar sensor (stock, up to 1.45 bar of boost control), 3.0 bar (up to 1.95 bar), 3.5 bar (up to 2.45 bar), 4.0 bar (up to 2.95 bar) or 5.0 bar. The map viewers show the pressure maps scaled for this sensor.

Turbo
Information field used as an indicator to TxSuite which type of turbo is mounted in the car. Please set to the correct type for good operation: Stock (T25 mostly), GT17, TD04-15T, TD04-19T, GT28BB, GT28RS, GT3071R (either .86 or .64), HX35w, HX40w or S400SX371.

Tuning stage
Information field, used by TxSuite to see whether the file has been tuned before or not.

Synchronization timestamp
Indicates when TxSuite has last altered the content of the maps in the file. This timestamp is used to determine whether to synchronize maps from the file to the ECU or the other way around.

Fuelling

Constant injection (E51)
Turns a constant injection time on/off.

Load control
Turns correction factor for load changes on/off.

Fuelcut in engine brake
Turns fuel cut off during engine braking on/off.

No fuelcut in R12
Turns the fuel cut off during engine braking on/off in reverse, 1st and 2nd gears.

Idle

Constant injection time
Turns a constant injection duration during idle on/off.

Load buffering during idle
Turns load buffering during idle on/off.

Higher idle during start
Turns a higher idle rpm during starting on/off.

Use idle injection map
Turns the usage of a separate injection map during idle situation on/off.

Idle control
Turns separate idle control on/off.

Misc

Airpump control
Turns the air pump control (SAI models) on/off.

Knock detection OFF
Turns off the knock detection algorithms in Trionic 5. Use with caution.

Purge control
Turns the purging function on/off.

Temperature correction
Determines whether the system should compensate for temperature differences or not.

Boost control
Turns the boost control valve control (APC) on/off.

Fixed idle ignition gear 1&2
Turns the usage of a fixed, constant ignition angle in first and second gear on/off.

Purge valve MY94
Turns the specific purge function for MY94 valves on/off.

Temp. corr. in closed loop
Turns the temperature correction in closed loop operation on/off.

Enrichment

Acceleration enrichment
Turns mixture enrichment during acceleration on/off.

Enrichment during start
Turns mixture enrichment during cranking on/off.

Deceleration enleanment
Turns mixture enleanment during deceleration on/off.

Enrichment after start
Turns mixture enrichment after starting the engine on/off.

WOT enrichment
Turns mixture enrichment during Wide Open Throttle situations on/off.

Adaption

Adaption of idle control
Turns adaptive behavior for idle control on/off.

Fuel adjustment during idle
Turns fuel adaption during idle state on/off.

Adaptivity
Turns fuel adaption on/off (spot adaption?).

Global adaption
Turns adaption on/off on a global level.

Adaptivity with closed throttle
Turns adaption with closed throttle plate on/off.

ECU specifics

CPU speed
Information field showing the CPU speed this bin was designed to run on.

Partnumber
The partnumber of the software. Normally this matches the number on the ECU housing. The magnifier button looks it up as File → Lookup partnumber does.

Dataname
Information field showing the software identifier called data name.

Software ID
Information field showing the software identifier for this specific binary file.

VSS code and VSS enabled
Turns VSS (Vehicle Security System) on/off and the VSS code to use for this car (if VSS is active).

Trionic 5.5
Information field showing whether this bin is a T5.5 binary or a T5.2 binary.

RAM locked
Toggles the RAM lock flag, red while it's on. If RAM is locked, no data can be written to it, so no live tuning is possible. If you want to alter maps and values in realtime RAM needs to be unlocked.

Tank pressure diagnostics
Turns fuel tank pressure diagnostics on/off.

Lambda control

Correction for engaging A/C
Determines whether an extra correction factor should be applied to the closed loop operation when the airco is engaged. This allows for other AFRs than 14.7 to be targeted when engine load is higher due to the airco components being activated.

Correction for TPS opening
Determines whether an extra correction factor should be applied to the closed loop operation when the throttle plate is opening. Opening the throttle plate results in an extra air rush, while the fuelling particles cannot keep up with this sudden airflow change resulting in a lean mixture. This option allows for other than 14.7 AFRs to be targeted in these conditions.

Lambda control
Turns closed loop operation on/off. If closed loop is activated the system will be in closed loop when certain criteria are met. If closed loop is deactivated here, the system will never enter closed loop and mixture will be determined by fuelling settings only.

Lambda control during idle
Determines whether closed loop operation should be active during idle conditions.

Lambda control on transients
Determines whether closed loop operation should stay active even when transients occur in the engine load or throttle plate opening.

Enable second lambda sensor
If your car is fitted with two lambda sensors (one pre-cat and one post-cat sensor) this should be enabled. Most models only have one lambda sensor and therefore should have this option turned off.

Advanced

The Advanced button shows a few more fields: the hardcoded RPM limit, Interpolation of delay, Throttle acc/ret adjust MY95 and Constant angle. If you don't know what they do, leave them alone.

Trionic 5 tools

The last section of the Actions menu has the tools only a T5 has.

Actions → Examine binary

This option will run an examination process on the opened binary file and TxSuite will generate a report of its findings. This examination is not 100% fool-proof but it will give you a pretty accurate idea on what an unknown binary file was intended to be used for hardware wise.

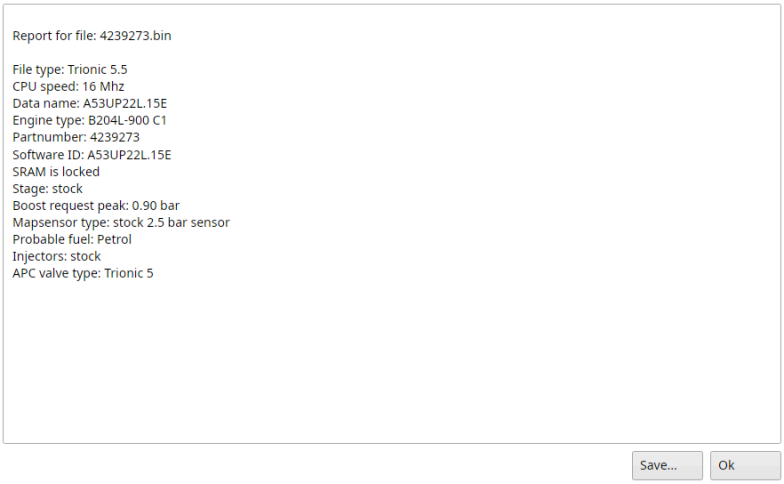


Image 28: Actions → Examine binary on a stock T5 binary

The report shows in a list, Save stores it as a text file.

Actions → Check for anomalies

Builds a report of anomalies (things that should normally not occur) that TxSuite finds in the currently opened binary file.

Actions → Open a saved report

Lets you review previously saved reports. Reports are text files now, T5Suite's old report format can't be opened.

Actions → Show dyno graph

This will show you a rough indication of what your dyno graph should look like based on the current binary: torque, power and injector duty cycle. The result of this graph largely depends on whether the boost request can be met. Refresh draws it again after you changed a map, Export saves the graph as a PNG picture.

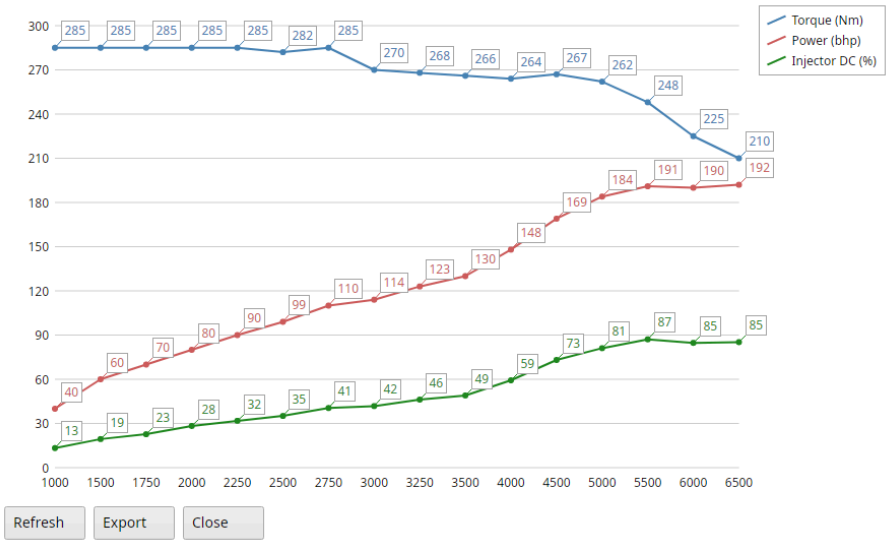


Image 29: Actions → Show dyno graph on a stock T5 binary

Actions → Show compressor map

This will show you a rough indication of where you will land in the compressor map with your current boost request. This of course assumes that you can actually reach your requested boost. TxSuite guesses the turbo from the file's footer and partnumber, you can pick another and set the volumetric efficiency.

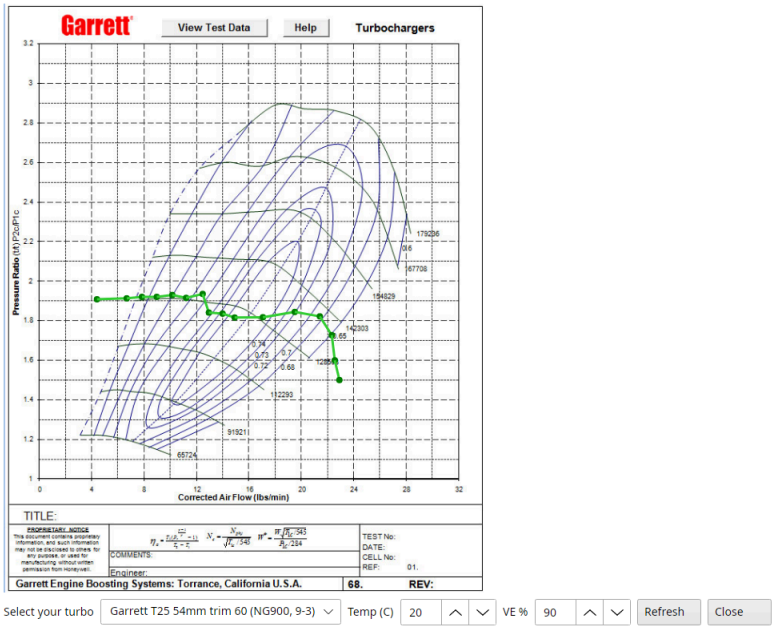


Image 30: Actions → Show compressor map on a stock T5 binary

Actions → Injection timing viewer

This starts a new screen showing the approximate injection duration figures for each engine speed and load point. It shows milliseconds or duty cycle for each cell of the fuel, idle or knock fuel map,

at an intake temperature, battery voltage and injector constant you set. You can use this data to make sure your injectors are big enough and small enough ;)

	-0.78	-0.70	-0.62	-0.52	-0.44	-0.36	-0.28	-0.20	-0.04	0.12	0.28	0.44	0.60	0.76	0.92	1.08
6200	1.81	2.18	2.76	3.34	4.08	4.81	5.53	6.43	7.89	9.59	11.36	13.57	16.37	18.1	20.44	22.91
5820	1.8	2.27	2.82	3.53	4.28	4.97	5.69	6.53	8.07	9.7	11.48	13.75	16.13	18.37	20.74	23.17
5440	1.85	2.32	2.86	3.55	4.27	4.95	5.62	6.46	8.01	9.7	11.44	13.44	15.69	18.26	20.26	22.65
5060	1.89	2.36	2.92	3.68	4.39	5.04	5.67	6.36	7.83	9.49	11.25	13.13	15.29	17.72	20.09	22.65
4680	1.84	2.3	2.91	3.7	4.37	5	5.67	6.26	7.56	9.08	10.69	12.47	14.41	16.91	19.26	21.63
4300	1.84	2.3	2.9	3.64	4.28	4.93	5.6	6.26	7.53	9.01	10.54	12.16	13.96	16.69	19.2	21.5
3920	1.85	2.31	2.88	3.58	4.22	4.89	5.51	6.14	7.45	8.9	10.26	11.67	13.47	15.66	17.6	20.22
3540	1.84	2.29	2.86	3.61	4.23	4.83	5.44	6.09	7.27	8.66	10.18	11.58	12.98	14.85	17.13	19.71
3160	1.83	2.27	2.82	3.53	4.15	4.75	5.38	5.99	7.12	8.42	9.87	11.23	12.54	14.14	16.19	18.56
2780	1.81	2.23	2.72	3.39	3.99	4.59	5.2	5.79	6.94	8.25	9.47	10.74	12.19	13.71	15.42	18.05
2400	1.8	2.24	2.76	3.43	4.03	4.65	5.27	5.89	7	8.21	9.47	10.69	11.94	13.33	14.71	17.02
2020	1.8	2.22	2.7	3.31	3.89	4.47	5.11	5.74	6.83	7.97	9.16	10.34	11.5	12.79	14.3	16.96
1830	1.82	2.25	2.76	3.4	3.96	4.51	5.11	5.74	6.83	7.94	9.12	10.34	11.5	12.85	14.42	16.96
1640	1.81	2.26	2.78	3.44	3.99	4.51	5.09	5.69	6.83	7.97	9.12	10.38	11.7	13.06	14.42	15.49
1260	1.81	2.23	2.71	3.31	3.84	4.39	4.96	5.5	6.56	7.77	9.16	10.61	12.09	13.22	14.36	15.49
900	1.81	2.18	2.56	3.05	3.61	4.45	5.2	5.67	6.74	8.01	9.63	11.23	12.49	13.66	14.83	16

Intake Air Temperature 60 °C Condition Normal Battery voltage 13 V Injector constant 20 Show value for Injection duration Close

Image 31: Actions → Injection timing viewer on a stock T5 binary

Actions → Merge binary files

Only useful when you have two binary files (one for each flash chip in the ECU) and you want to stitch them together (advanced users).

Actions → Split binary file

Only useful when you want to have two binary files (one for each flash chip in the ECU) created from the current binary file (advanced users). They are written next to the bin as `chip1.bin` and `chip2.bin`, overwriting files with those names.

Tuning menu

The Tuning menu starts with the tuning wizards, the rest gives you direct access to the most important maps, grouped in submenus. Each map entry opens that map of the open file in a map viewer. An entry only shows when the binary has the map, so a T5.2 file has no gear limiters and no ignition retard limit. Point at an entry to see which symbol it opens.

Tuning → Tuning wizards

Tune me up ® is always there. The other wizards are advanced and only show with File → Settings → T5 → Advanced mode enabled. Tune me up, the MAP sensor conversions and Convert to E85 write a report of what they changed, Save… stores it as a text file.

Tune me up ®

Lets you convert your stock binary file to a stage I, II or III equivalent, or higher. Choose the desired tuning stage:

- Stage 1 (+30 bhp, +40Nm)
- Stage 2 (+50 bhp, +60Nm)
- Stage 3 (+80 bhp, +80Nm)
- Stage 4 and higher

A backup of your original file is created in the process before changes are made to it. A file already tuned to a higher stage isn't touched.

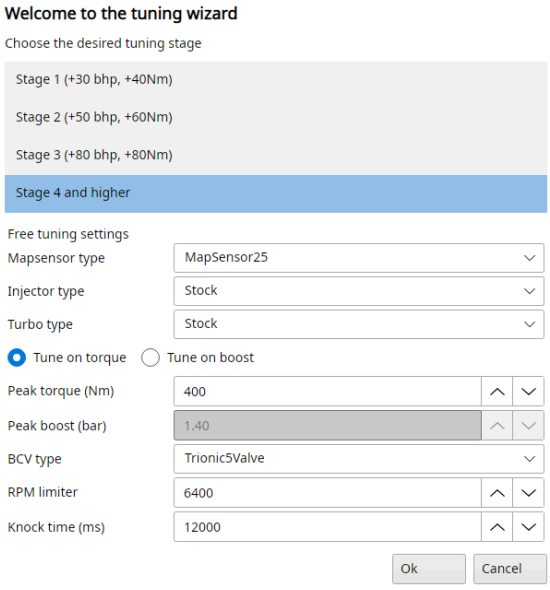


Image 32: Tune me up ® with Stage 4 and higher selected

Stage 4 and higher uses the free tuning settings: map sensor, injectors, turbo, tune on torque or on boost with the peak torque or peak boost, the boost control valve (BCV) type, the RPM limiter and the knock time. Depending on your input TxSuite will try to generate a close as possible match to your hardware. Please do note that tuning beyond stage III will require you to verify AFR settings and knock sensitivity in realtime to make sure everything is working properly. Mapping for cold- and warm-start are also known issues to address after tuning for larger injectors and/or different mapsensor types.

Convert to different MAP sensor

Lets you convert your binary file for usage with a 2.5, 3.0, 3.5, 4.0 or 5.0 bar mapsensor, a submenu item each. All boost related tables are altered so the ECU calculates the right values with the new sensor. Since this is an advanced topic you will understand that manual checks need to be done and changes need to be made for the file to be optimal for usage in your setup.

Convert to larger injectors

This will start a wizard which will lead you through the relevant maps and it will make proposals on what to change: choose your injector type, and it shows the current and proposed injector constant, battery correction map and cranking factor.

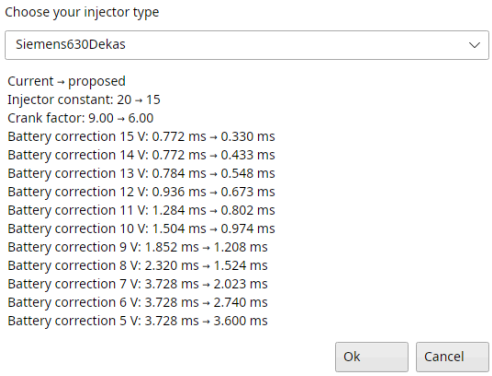


Image 33: The injector wizard with Siemens 630cc injectors chosen

Convert to E85 (ethanol) fuel

Lets you convert your binary file for usage with E85 fuel. Please note that Trionic 5 does not support bi-fuelling, once converted it will no longer run on regular petrol. Since this is an advanced topic you will understand that manual checks need to be done and changes need to be made for the file to be optimal for usage in your setup. Maps that are changed by the wizard are: main and warm-up ignition maps, global fuelling (injector constant), cranking- and after-start- enrichment. Running it twice enriches twice.

Change boost adaption ranges

For bigger turbos, the boost adaption ranges do not match up with what is hardcoded into the binary file (the stock ranges are optimized for T25/TD04 turbo types). You can adjust these ranges with this feature, the rpm range for manual and for automatic gearboxes, and the Max. boost error (bar) too. The hardcoded ranges will be altered in your binary file, this wizard adjusts the actual code!

Change boost bias range

In late binaries (probably all OBDII files) the boost bias map (Reg_kon_mat!) has hardcoded rpm support points (y-axis points). Since these support points run only up to 5500 rpm, this is not enough to support good control when using a large turbo which requires more control in higher rpm ranges as well. You can alter the code in the binary using this option and stretch the support points range. The axis step range sets it: the support points run from 2500 rpm up to 2500 + 300 × step rpm, 5500 rpm at the stock step of 10.

Change RPM limit

Sets the software RPM limit (Rpm_max!) and the limit hardcoded in the program, T5.5 only.

Tuning → Injection [Fuel]

VE map - normal
Lets you edit the main injection correction map directly (Insp_mat!). Please note the black rectangular indicators in the viewer, see Image 1. These indicate the cells that operate in closed loop. Blue indicators indicate the knock adaption range and white indicators show the boost adaption range.

VE map - knock
Lets you edit the knock injection correction map directly (Fuel_knock_mat!).

Injector scaling
Lets you edit the main injection constant directly (Inj_konst!).

Battery correction map
The injection time correction for battery voltage (Batt_korr_tab!).

Fuel cut in overboost
Lets you view and edit the fuel cut level (Tryck_vakt_tab!). This map is shared between manual and automatic cars. Note: 255 (or 1.55 bar in easy view) means the fuel cut function has been turned off.

Tuning → Ignition

Ignition normal
Lets you edit the main ignition map directly (Ign_map_0!).

Ignition knock
Lets you edit the ignition knock map (ignition pull values) directly (Ign_map_2!).

Ignition warmup
Lets you edit the warm-up ignition map directly (Ign_map_4!).

Tuning → Turbo control [M] and Turbo control [A]

The same maps for cars with a manual gearbox [M] and with an automatic gearbox [A].

Boost request map
Lets you view and edit the boost request values (Tryck_mat! and Tryck_mat_a!).

Boost control bias
Lets you view and edit the bias (pretension) values for the boost controller (Reg_kon_mat! and Reg_kon_mat_a!).

P factors, I factors and D factors
Lets you view and edit the PID values for the boost controller in Trionic 5. For more information on PID controller functioning please refer to the Trionic 5 documentation.

Boost limit in 1st gear and Boost limit in 2nd gear
Lets you view and edit the boost limiter values for 1st and 2nd gear (only 1st gear for automatic gearboxes). These limiters can be disabled.

Tuning → Knock detection

Knock sensitivity map
How sensitive the knock detection is (Knock_ref_matrix!).

Ignition retard limit
How far the ignition may be retarded on knock (Knock_lim_tab!), T5.5 only.

Boost reduction map
How much boost is taken away on knock (Apc_knock_tab!).

Tuning → Warmup

Afterstart enrichment (1) and (2)
Lets you view and edit the first and second after start enrichment factors. They depend on coolant temperature (y axis). These are multiplication factors on the final injection time.

Tuning → Idle

Idle target RPM
Lets you set the target idle engine speed values (depending on coolant temperature).

Idle ignition
Lets you set the default idle ignition advance value.

Idle ignition correction
Lets you set the ignition control for idle state. The target rpm value error (actual rpm - target rpm) is the y axis for this map.

Idle fuel map
Lets you set the fuelling control values for idle state.

My Maps menu

My Maps is your own menu of the maps you use most, so they are one click away. Each ECU has its own, the menu shows the open file's ECU's maps.

My Maps → Define myMaps...

Opens the list of your maps, a row per map with its category, title and symbol. Add and Remove change the list, Ok saves it. The first time the list starts with the old suite's defaults for that ECU.

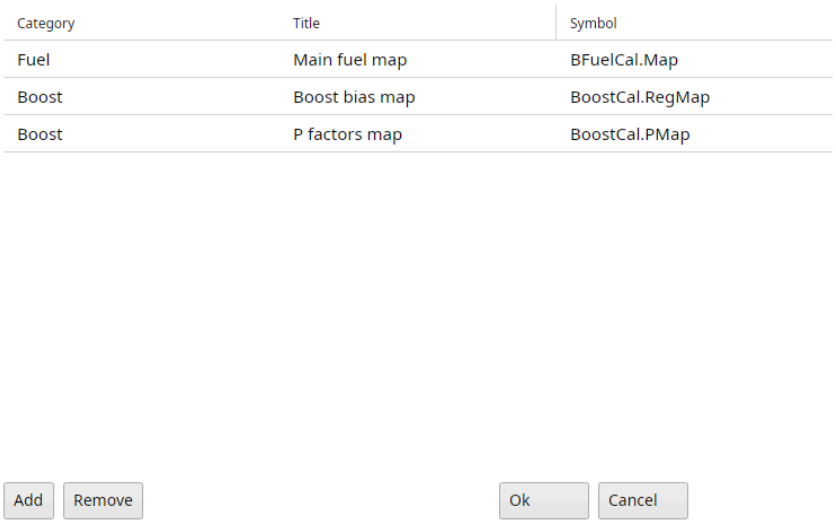


Image 34: My Maps → Define myMaps..., the T7 defaults

Below Define myMaps... the menu has a submenu per category, with its maps. A map entry opens that map of the open file.

The symbol menu's Add to MyMaps adds the selected symbol to the list.

The list is kept per ECU in `mymaps.xml` in TxSuite's settings folder, the first start takes over the old suites' lists.

Realtime menu

The Realtime menu connects to the ECU, runs the realtime panel and works with its logfiles. The ECU must be connected through one of the interfaces in Supported interfaces, set up in File → Settings → Common. A file of the ECU must be open, it tells TxSuite which ECU it's talking to.

Realtime sessions on a T5 only run over CAN (the P-bus), not over BDM!

The items only your ECU has are described in the next chapter.

Realtime → Connect ECU

Connects to the ECU, the item then reads Disconnect ECU. The status bar shows the connection. Map viewers opened while connected show the ECU's data, see Maps from the ECU in Working with maps.

If the ECU and the binary file are not in sync, TxSuite will ask you if you want to resynchronize, see Data synchronization in the next chapter.

Realtime → Toggle realtime panel

Allows you to start a realtime session [Shift+F1]. If all connections are done well, TxSuite should be able to connect to the ECU and show you realtime information. Toggling it again stops the panel.

While the panel runs it logs every value it reads into a logfile next to the bin, one per bin and day: `<bin>-yyyyMMdd-CanTraceExt` with the ECU's log extension.

The T5 realtime panel is described in the next chapter.

Realtime → Export logfile to LogWorks

Will convert and export a logfile to LogWorks format (`.dif`). You choose the symbols and the time range to export in "Logfile data selection". It lists every symbol of the log, all ticked, with Select all and Select none. From time and Up to time start at the beginning and the end of the log. TxSuite doesn't start LogWorks, open the file in it yourself.

Realtime → Load logfile

Loads and shows a logfile in TxSuite's log viewer.

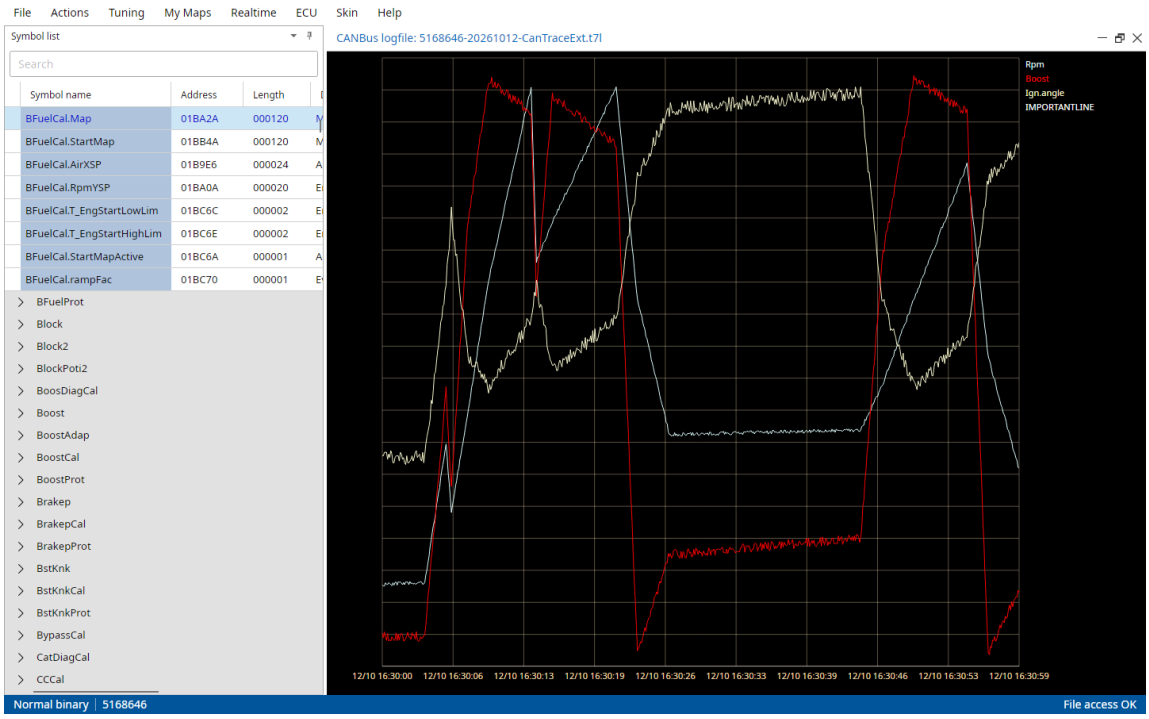


Image 35: A T7 log in the log viewer

A log is split into sections where it has a gap of more than 10 seconds, you pick the section first. The channels are drawn over each other, each scaled to its own range, in the colours of Set symbol colors. The legend shows the values under the cursor. The mouse wheel zooms, dragging moves along, and the arrow keys do the same.

Realtime → View matrix from logfile

Lets you view three channels in a logfile displayed as a surface graph (matrix). Choose the symbols for the x axis, the y axis and the value, and whether a cell gets the mean, the minimum or the maximum of its values. The result is a 16 × 16 map between the smallest and largest logged x and y, in a read-only map viewer with its 3D graph.

Realtime → Set symbol colors

Lets you choose the colours that will be used for the symbols in the log viewer. The list has every symbol the ECU has in RAM, with a search box on top.

Realtime → Get fault codes

Reads all the error code counters in the ECU and displays which error counter holds values greater than zero.

Clear clears the selected fault code, Close closes the window.

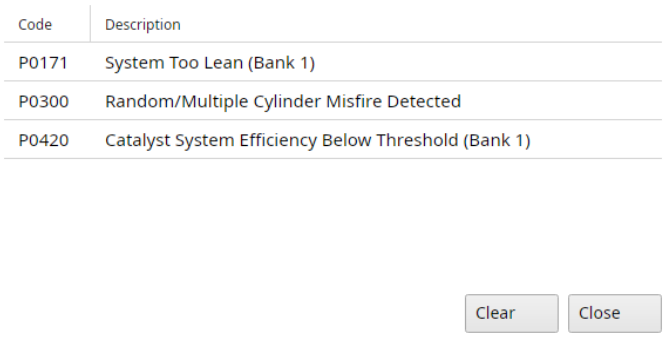


Image 36: Fault codes

Realtime → Write log marker

While running a realtime session this enables you to write a marker in the log which you can track back easily later on [F6].

Realtime → Export logfile to CSV

Exports a logfile to CSV, with the symbols and the time range you choose in “Logfile data selection”, as for LogWorks. The first column is the time in seconds since the first line.

Trionic 5 realtime

The Realtime menu's items only a T5 has, the T5 realtime panel and Autotune.

Once connected, TxSuite works on the ECU and the file together. Map viewers show the SRAM content of the ECU, and saving a map writes it into the ECU and the file in one go.

Data synchronization

When you connect, TxSuite compares when the file and the ECU were last changed. If the ECU and the binary file are not in sync, it shows "Data synchronization" with both timestamps and the proposed direction, ECU to binary or binary to ECU.

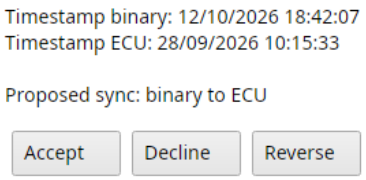


Image 37: Data synchronization, the file newer than the ECU

Accept
Synchronizes in the proposed direction.

Decline
Does nothing.

Reverse
Synchronizes the other way around.

Synchronizing reads every map from the ECU and compares it with the file, that takes a while. The status bar shows "Sync: N%" and then "Synchronized".

Realtime → Synchronize maps

Lets you synchronize maps between the binary file and the connected ECU, in the direction the timestamps tell.

Realtime → Compare ECU with binary

This option checks all differences between the ECU and the binary you've loaded. It takes an SRAM snapshot first, next to the bin, and shows the differences as Actions → Compare binary to SRAM snapshot does.

Realtime → Configure realtime panel

Allows you to configure the realtime panel in offline mode. This way you can pre-configure the panel before even starting your car. Realtime → Toggle realtime panel then starts it.

The realtime panel

Once you've started the realtime panel and the canbus connection is running, it should look something like this:

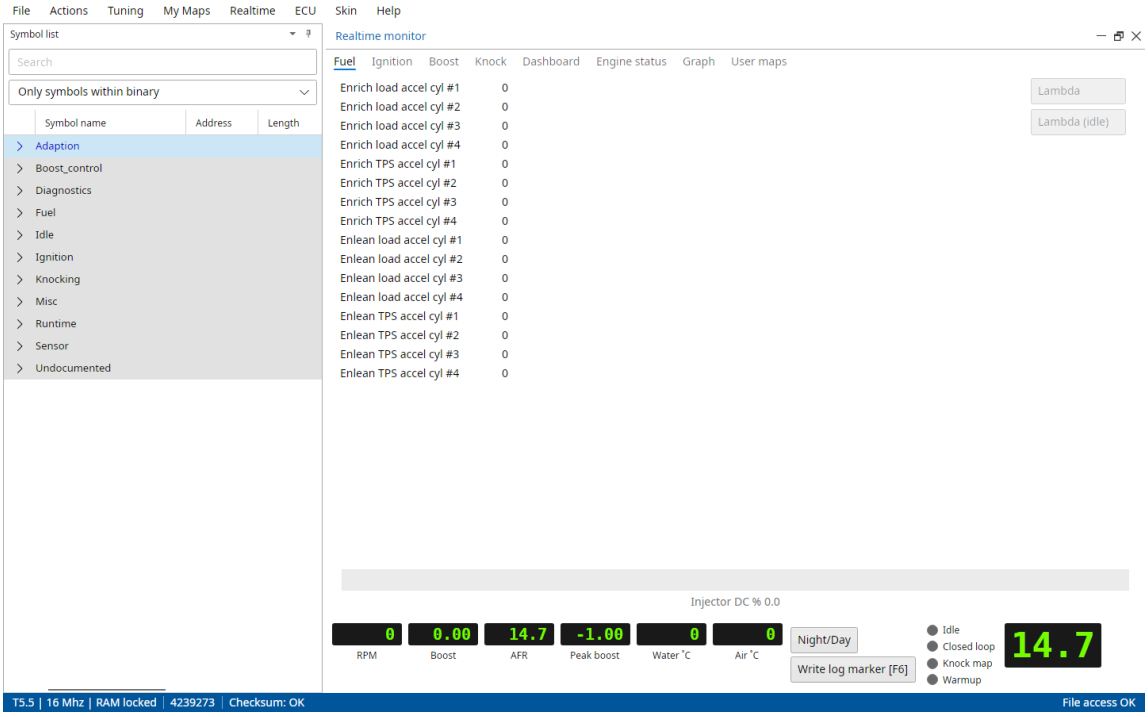


Image 38: The realtime panel, Fuel tab

The strip at the top is on every tab: rpm, temperatures, AFR (click it for lambda), boost and peak boost, and the LED's Idle, Closed loop, Knock map and Warmup. Night/Day switches the panel to dark colours, Write log marker [F6] writes a marker into the log.

You can watch and edit maps which hold a relation to tuning boost in the Boost tab. The Ignition tab will get you access to ignition related maps etc. You can directly start a map viewer by clicking the "Edit maps" button, which offers the maps of that tab.

Fuel, Ignition, Boost and Knock
: The values that matter for each, with switches for Lambda, Lambda (idle), APC control and Knock control.

Dashboard
: Speed, torque, power and their peaks, ignition and consumption.

Settings
: Switches for the program mode (Pgm_mod!) flags in the ECU, read back after every change.

User defined
: Your own list of symbols, as the free logging list of the other ECU's. The symbol menu's Add to realtime list adds a symbol from the symbol list.

Autotune and Autotune ignition
: The autotune sessions, see below.

Engine status
: The 40 status bits of Pgm_status as LED's.

Graph
: The values over time, right-click to choose the lines.

User maps
: Maps you added with the symbol menu's Add to realtime user maps.

Settings, User defined, Autotune and Autotune ignition only show with File → Settings → T5 → Advanced mode enabled. Each tab only reads the values it shows, so the panel stays fast.

Realtime → Clear knock counters

Allows you to reset all the knock counters in the ECU to zero.

Realtime → Knock map snapshots

With File → Settings → T5 → Knock counter snapshot after disconnect TxSuite saves the ECU's knock counter map every time you disconnect or stop the realtime panel (T5.5). This lists them with their total number of knocks. Ok shows one, Compare shows the difference between two.

Realtime → AFR maps

Show AFR target map
: Lets you set your desired AFR values for each load and rpm point for this binary file. After logging with a wideband lambda sensor you can compare the target values to the feedback values.

Show AFR feedback map
: Shows you the measured average values from the realtime sessions you have done for this binary file.

Show AFR error map
: Shows you the difference between the target AFR values you have entered (or the default ones) and the measured feedback values.

Show idle AFR target map, Show idle AFR feedback map and Show idle AFR error map
: The same for the idle fuel map.

Generate AFR target map
: Lets you generate a new default AFR target map based on certain properties in the binary. If you use big injectors, the target for low rpm and low load will be richer for example.

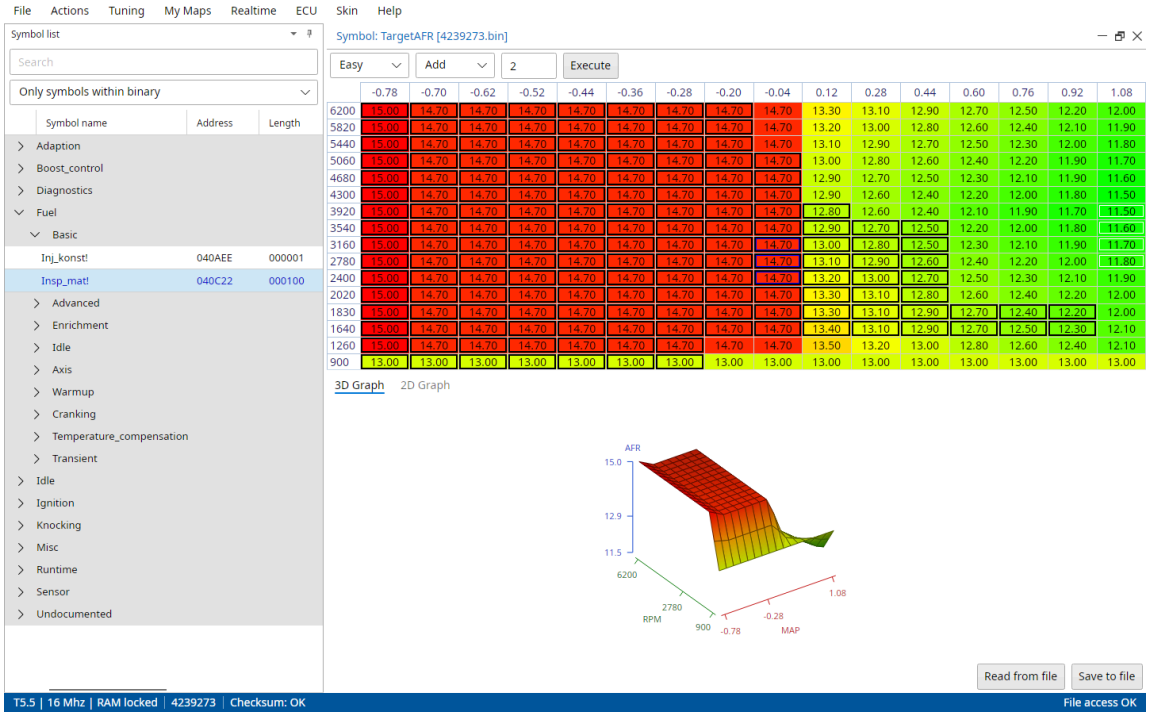


Image 39: Realtime → AFR maps → Show AFR target map

The maps are kept in an `AFRMaps` folder next to the file.

Realtime → Toggle Autotune

Starts and stops Autotune fuel [Shift+F5], the same as the Autotune fuel button in the panel. Advanced mode only.

Autotune criteria

For Autotune to become available you will have to meet certain criteria.

- You will have to be in advanced mode
- File → Settings → T5 → Use wideband lambda through symbol must be on, a wideband on a com port doesn't count
- Engine temperature should be over 70 degrees Celsius
- RAM should be unlocked, or the ECU won't take the changes

The Autotune fuel button shows in advanced mode. Click it and the session starts, or TxSuite tells you what's missing. The tab is then switched to "Autotune". Here you can see which cell in the fuel map is active and which cells are measured by Autotune with what AFR value. The moment Autotune is activated the software switches off closed loop control in the ECU by altering the Pgm_mod! symbol accordingly. This also switches off purge control because purging will occur only when the system is running in closed loop. While running the Autotune algorithm constantly checks whether all runtime criteria for making adjustments are met. These criteria are:

- No enrichment after fuelcut is allowed
- System is not allowed to run in the idle map, unless Allow idle map autotune is on
- Purge is not active
- Cooling water temperature enrichment is finished
- Afterstart enrichment is finished
- Closed loop control is inactive
- Engine should be warmed up
- Fuelcut is not active
- Knock map is not active

Optionally when Discard AFR measurements with fuelcut is active in the Autotune settings:

- Cylinder 1-4 should not be in fuelcut mode

Optionally when Discard closed throttle measurements is active in the Autotune settings:

- Throttle plate is not allowed to be fully closed

If all runtime criteria are met the algorithm measures AFR values through the wideband lambda input and determines the cell in which it should average this value by means of current engine speed and manifold pressure. Depending on the mode in which Autotune is running (direct update or user approve) it updates the fuel map in the ECU or it stores the averaged value in memory. In the latter case it shows the user a proposal for adjustments to be made after the Autotune session is ended.

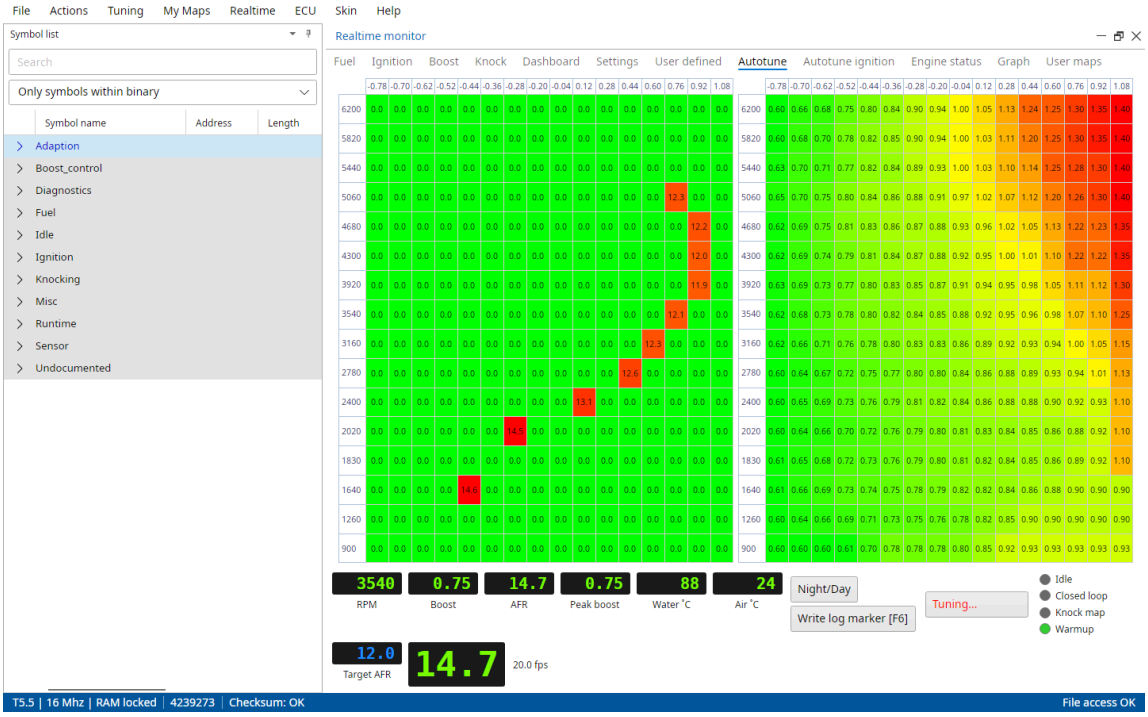


Image 40: The Autotune tab during a session: feedback AFR left, the fuel map being tuned right

After the Autotune session, the software will show you a map in which the cells to be corrected are highlighted and showing the percentages to correct the fuel map.



Image 41: Select percent mutations to accept

You can cancel this session, accept the selected cells (Accept selected, after selecting the cells you want to correct) or accept all suggested corrections (Accept all). TxSuite will save all mutated data into the ECU and the file.

With Auto update fuel map on, TxSuite asks “Keep adjusted fuel map?” when the session stops, No puts the original map back. Closed loop is switched back on when the session stops, also when you disconnect or flash in the middle of it.

Realtime → Autotune ignition

Starts and stops Autotune ignition, T5.5 and advanced mode only. It needs the realtime panel running and knock control on, the Knock control switch on the Knock or Settings tab. It advances the cells of the ignition map (Ign_map_0!) in the ECU step by step while the engine runs without knock, and retards and locks a cell when it knocks. Cells below the knock pressure limit aren't touched. The steps and limits are in File → Settings → T5 → Autotune settings. When the session stops TxSuite asks "Keep adjusted ignition map?", No puts the original map back.

Realtime → Ignition lock map

Opens the ignition map with a padlock on every cell Autotune ignition locked.

Realtime → Release locked ignition cells

Unlocks every cell, so the next Autotune ignition session can advance them again.

ECU menu

The ECU menu reads and flashes the ECU over CAN, the way the Trionic CAN Flasher does. Like the Realtime menu it needs a file of the ECU open, and an interface set up in File → Settings → Common. The progress shows in the status bar, and TxSuite won't close while it runs.

ECU → Read ECU

Downloads the flash content from the connected ECU. It will ask you where to store the binary file.

ECU → Flash current file to ECU

Flashes the open file to the connected ECU. Map viewers with unsaved changes are pointed out first, they are not in the file yet. A wrong checksum is corrected first, or TxSuite offers to correct it when File → Settings → Common → Auto update checksum is off. When it's done TxSuite says whether it worked.

A good check after flashing is to read the ECU back and compare the two files with Actions → Compare binary with other binary.

ECU → Get SRAM snapshot

Lets you create an SRAM snapshot from the ECU, a copy of its working memory, for later analysis or reference. It needs a connection to the ECU, Realtime → Connect ECU. In a project it goes into the project's Snapshots folder, otherwise TxSuite asks where to save it, proposing `Snapshot-<bin>-<time>.RAM`.

ECU → Upload SRAM to ECU

This option allows you to upload a snapshot of the ECU's SRAM (working memory) back into the ECU. This works like a sort of system restore point you can make. Connect first with Realtime → Connect ECU, without a connection nothing happens. TxSuite asks whether the maps should go into the binary file too.

Skin menu

TxSuite has a light and a dark skin.

Skin → Light, Dark and Follow the system

Light and Dark pick the skin, Follow the system takes the one your system uses. The choice is the same for every ECU and is remembered.

Skin → Cascade windows and Tile windows

Cascade windows puts the windows in the workspace on top of each other, slightly shifted, so every title bar shows. Tile windows horizontally and Tile windows vertically put them next to each other so they all fit.

Skin → Dock all floating windows

Brings every window that floats outside the main window back into the workspace.

Help menu

The Help menu has the manuals and documentation of all three ECU's, whatever file is open.

Help → T5, T7 and T8

Each has a submenu with the User manual, this book for that ECU, and the Trionic N documentation, Dilemma's book about that ECU.

Help → Release notes

Opens the releases page with what changed in each version.

Help → Check for updates

Looks for a newer TxSuite and shows the result in the status bar. When there is one it asks, OK downloads the setup on Windows and opens the release page elsewhere. TxSuite also checks when it starts.

Help → About TxSuite

The version, and the people who made the suites possible. TxSuite was created with the help of lots of people on ecuproject.com and trionictuning.com.